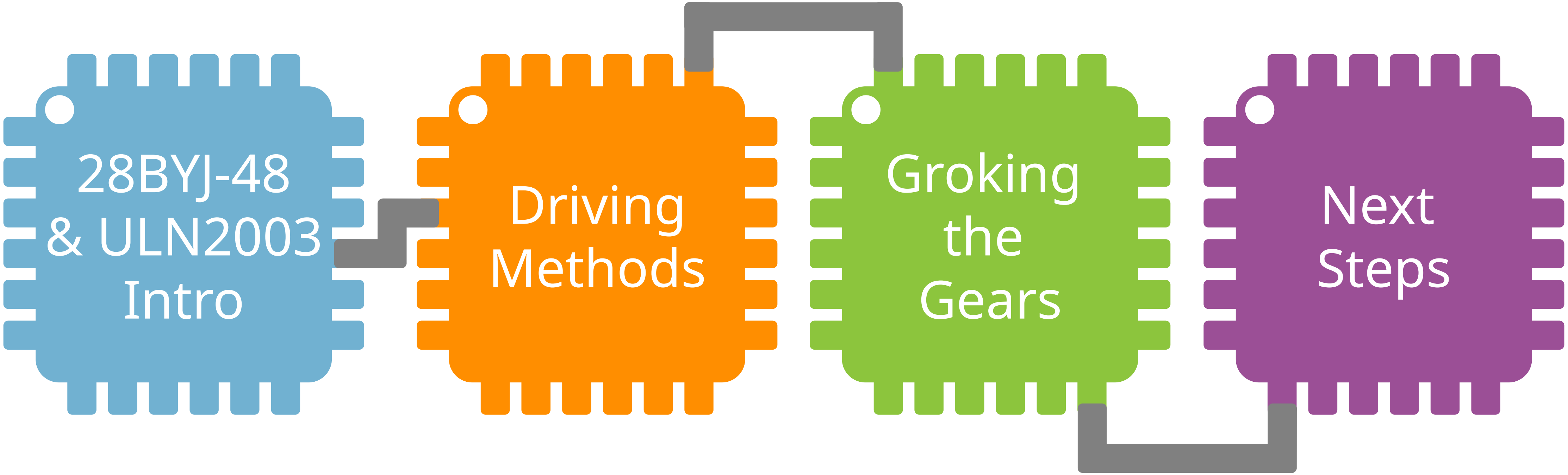
THE 28BYJ-48 STEPPER MOTOR AND THE ULN2003 DRIVER

Adapted from slides by Bret Stateham

(Microsoft)

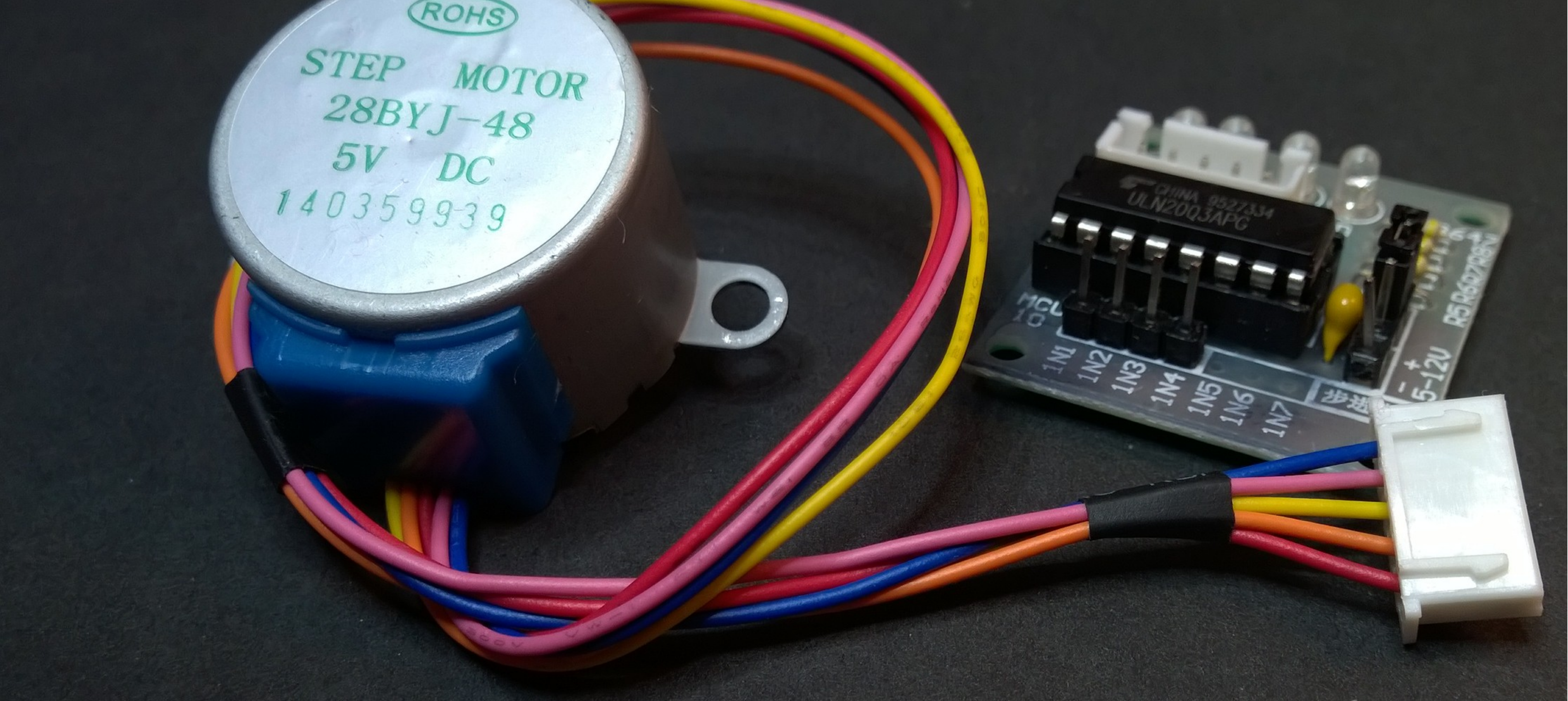
Microsoft

OVERVIEW

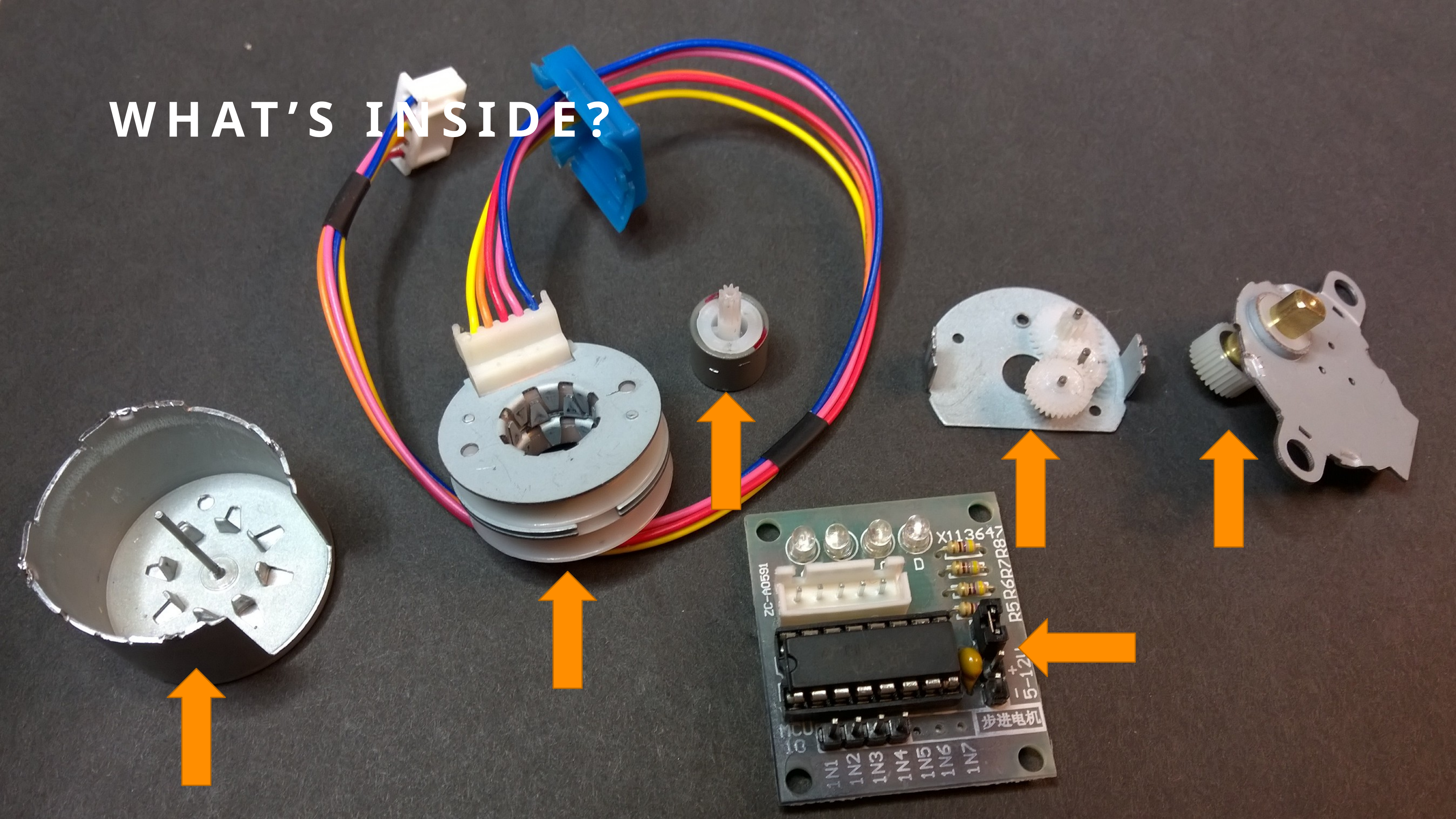

28BYJ-48 & ULN2003 Intro
Driving Methods
Groking the Gears
Next Steps

28BYJ-48 & ULN2003 INTRO

THE 28BYJ-48 MOTOR AND ULN2003 DRIVER…

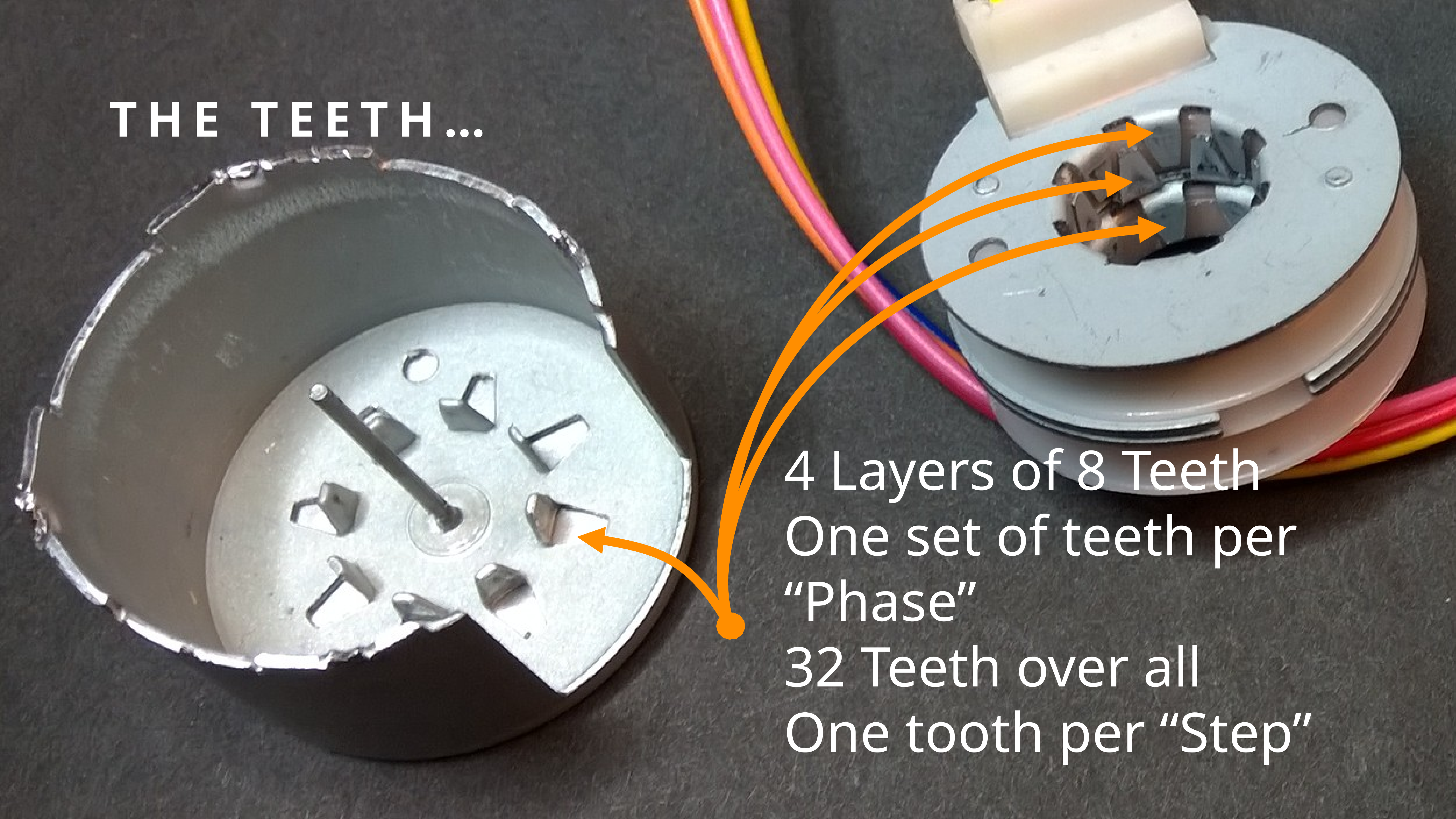


WHAT'S INSIDE?



THE COILS…

2 Coils, each in a “Unipolar” configuration with “Center Taps”

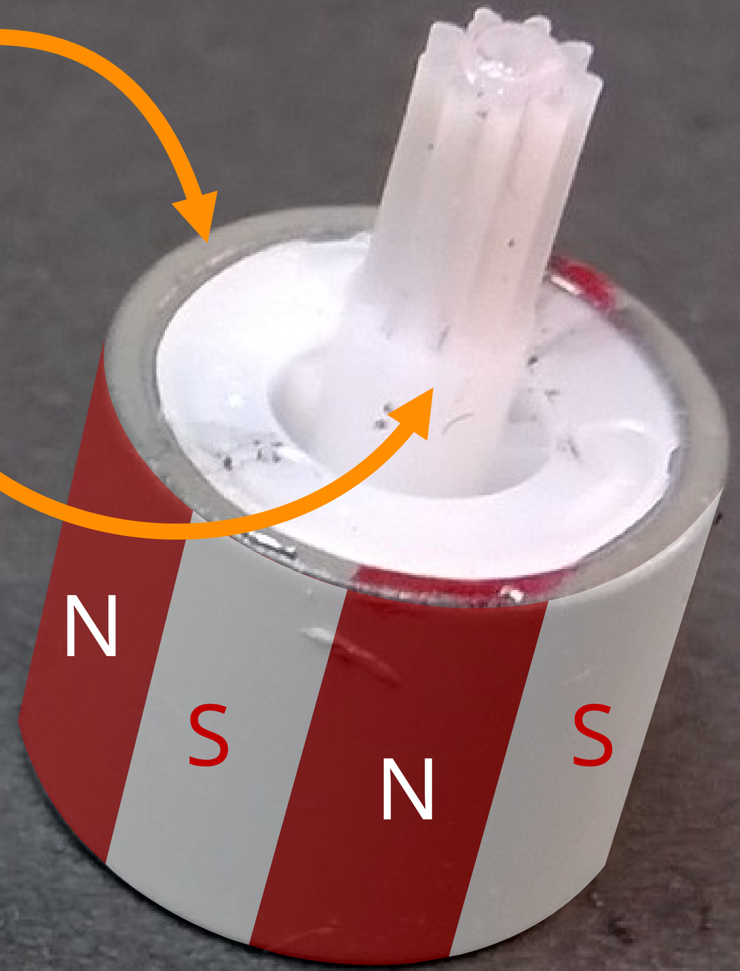

THE TEETH...
4 Layers of 8 Teeth
One set of teeth per
“Phase”
32 Teeth over all
One tooth per “Step”

THE ROTOR…

Permanent Magnet on a plastic body

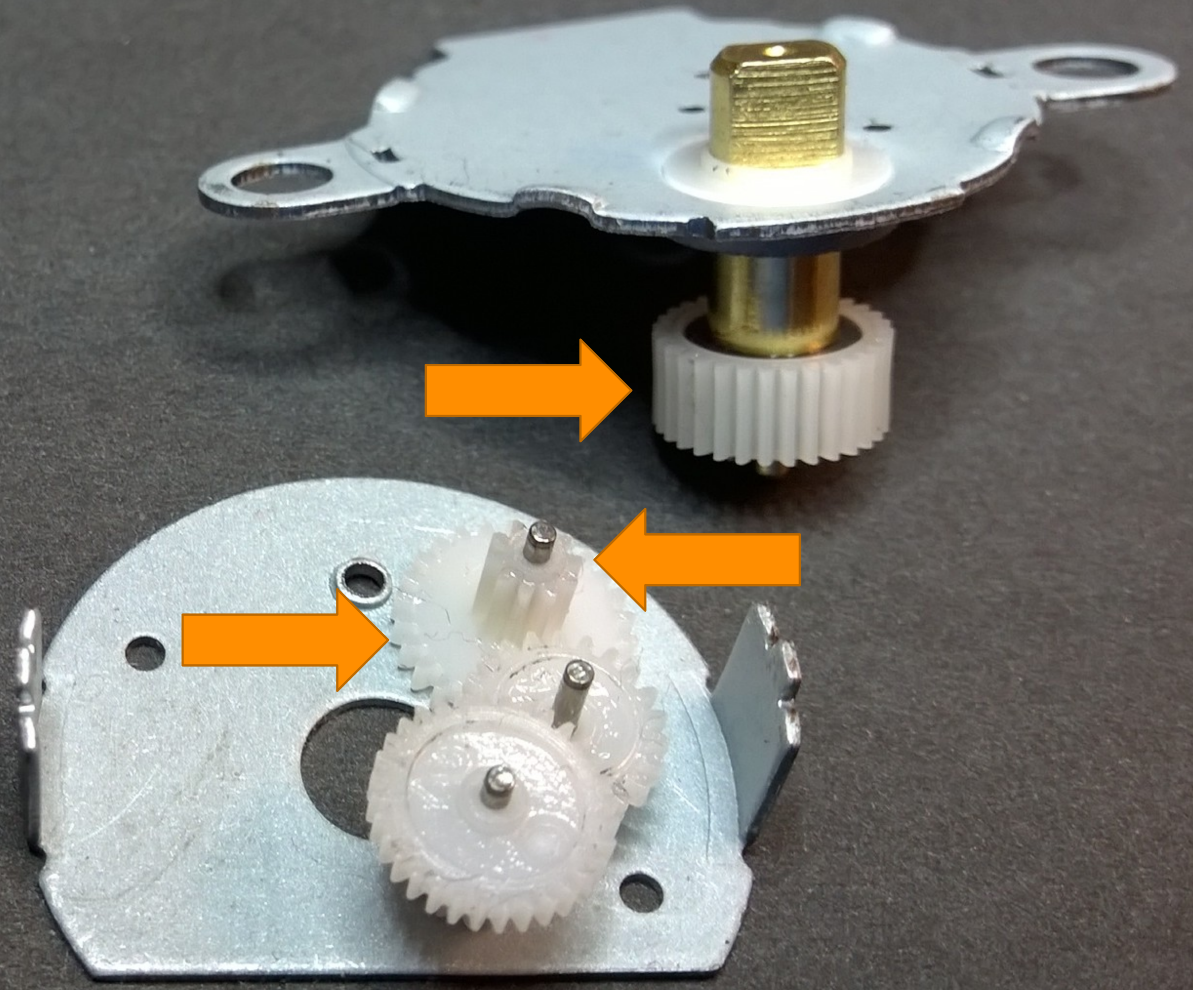
The shaft is a gear with nine teeth.

The rotor slides down on a post extending from the bottom of the motor case & gets surrounded by the coils and teeth



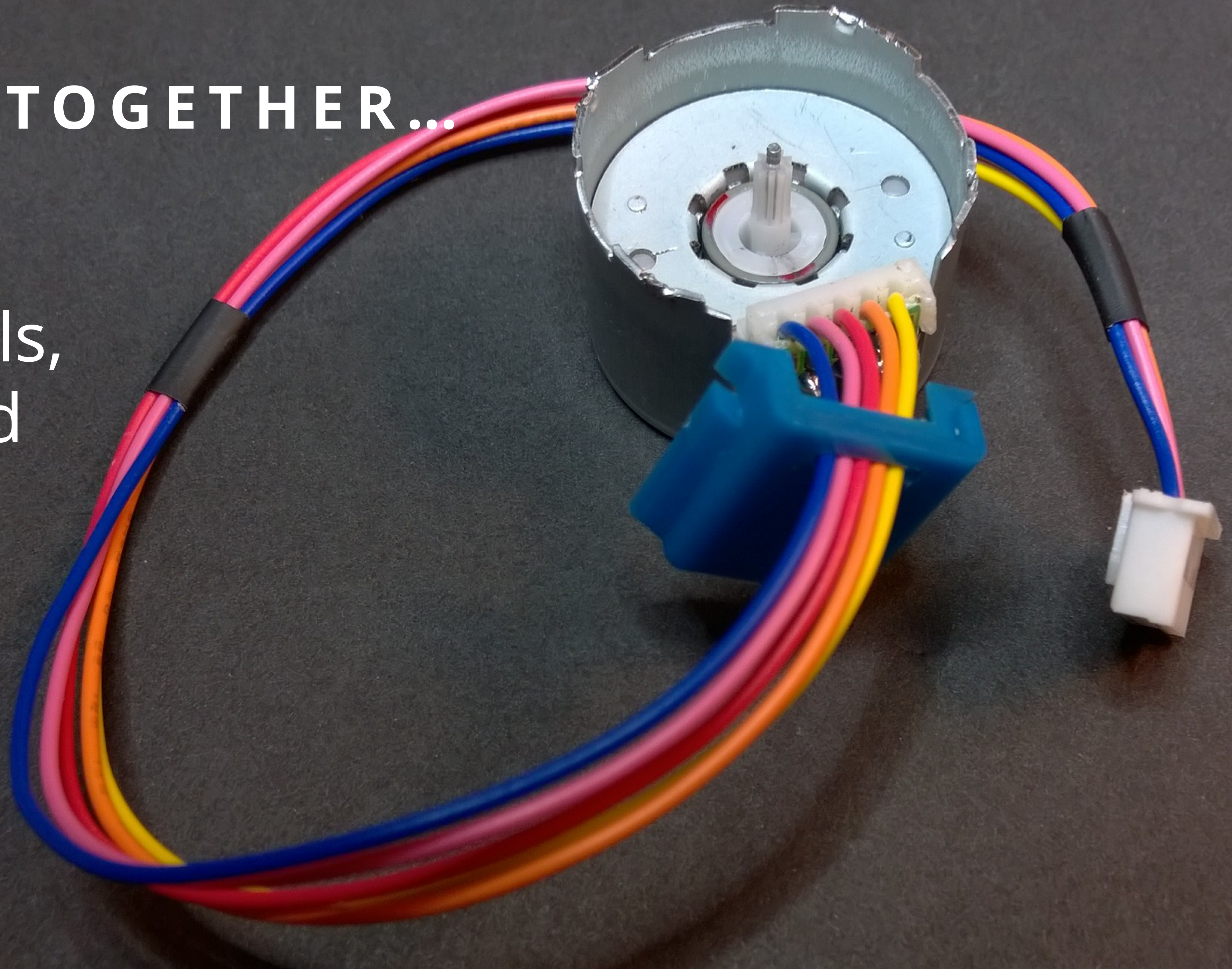
THE GEARS & SHAFT…

Overall 64:1 Gear Ratio. That means the motor rotor must turn sixty four full rotations for the outer shaft to turn once.



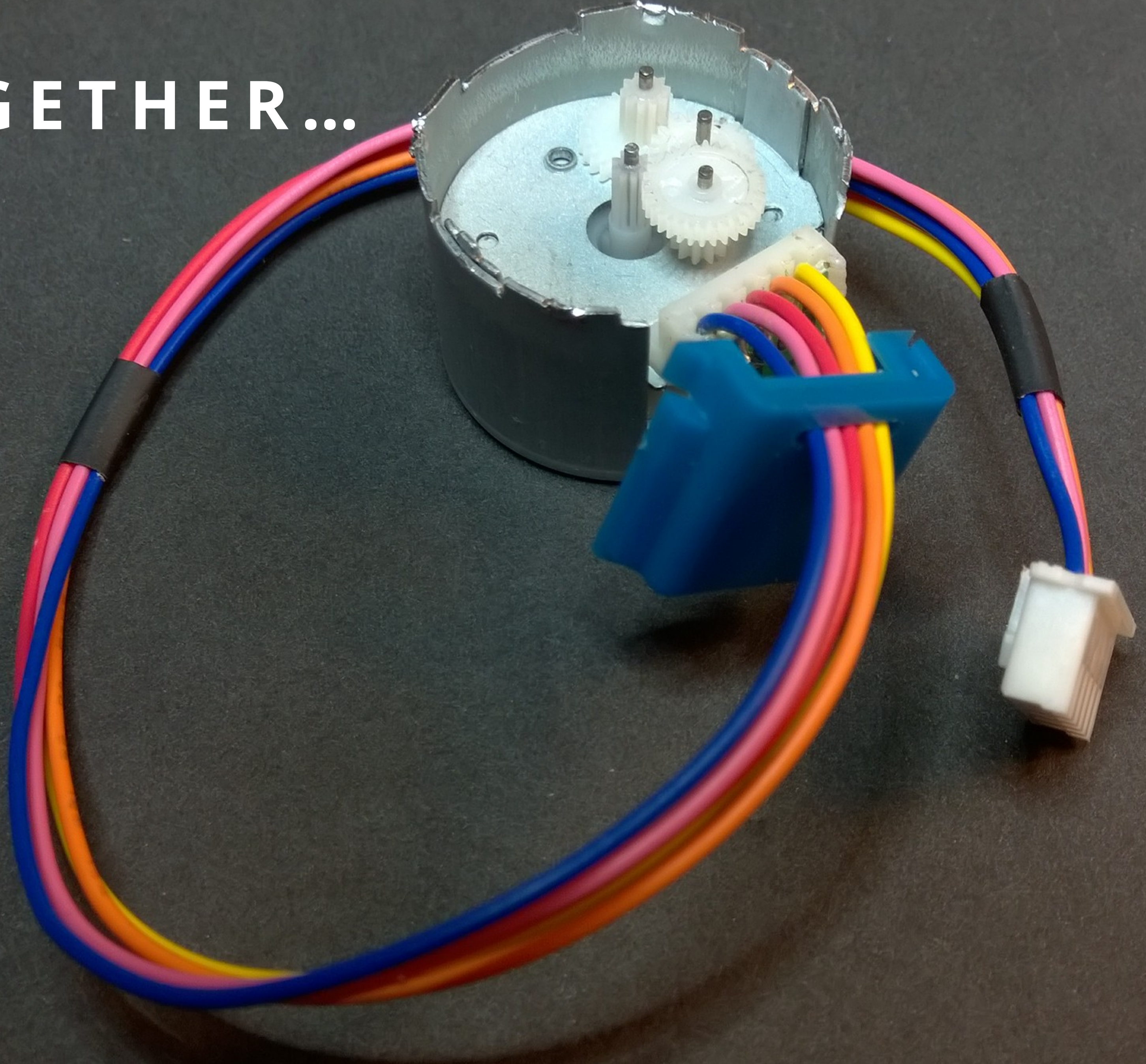
BACK TOGETHER…

First the Case, Coils, Teeth and Rotor…



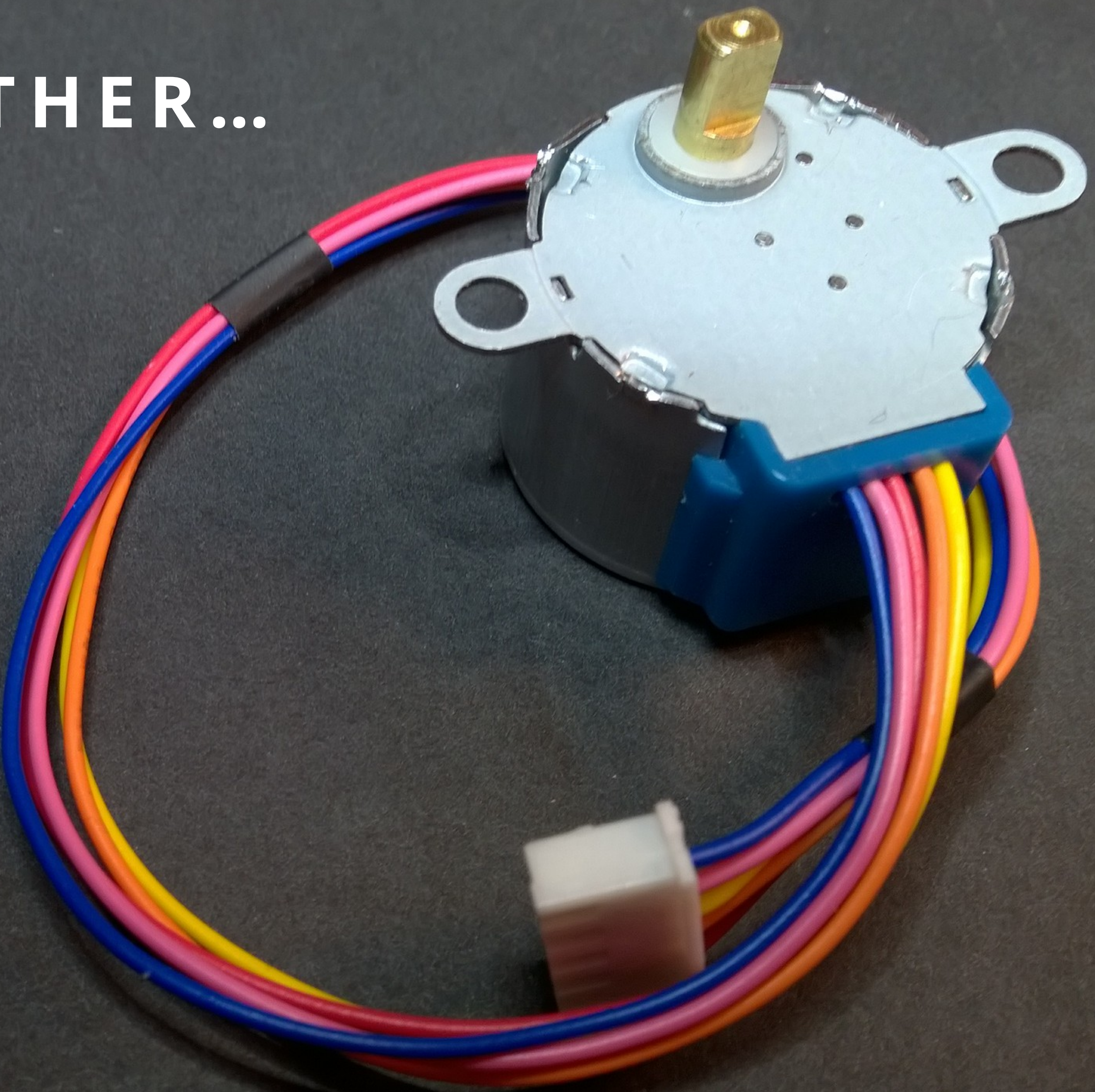
BACK TOGETHER…

Then the Gears…

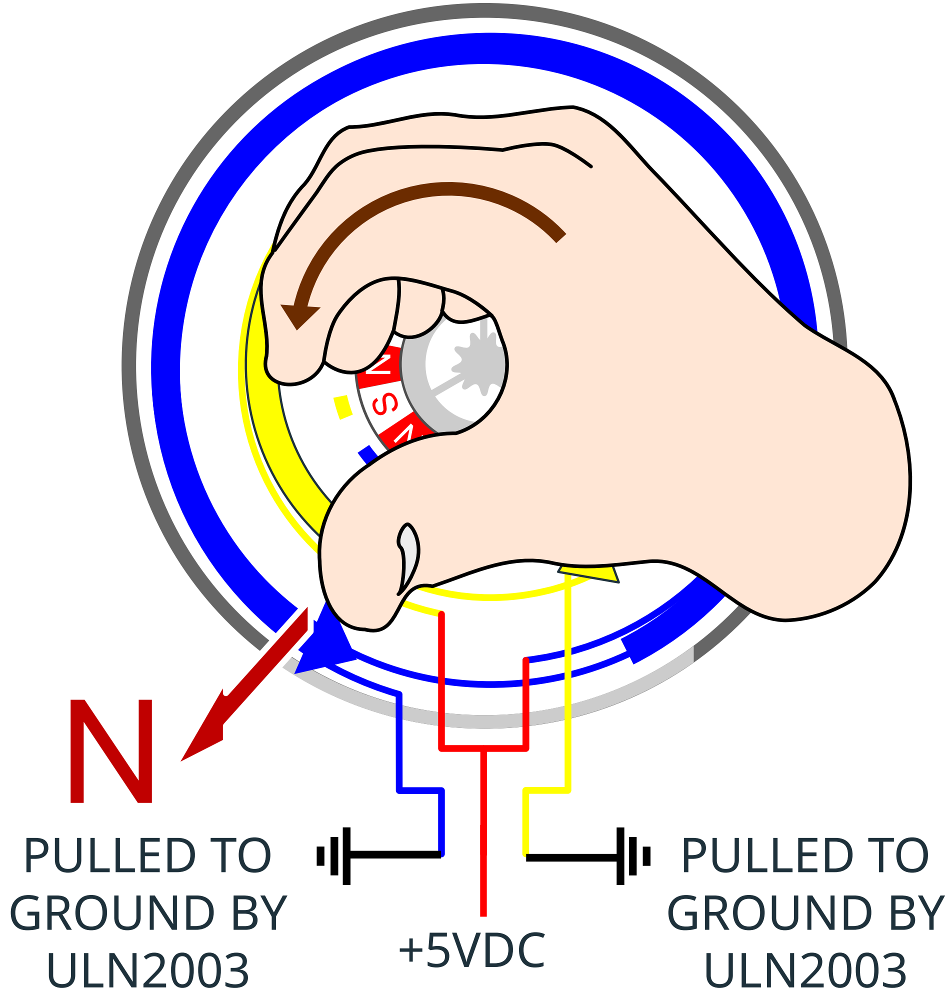


BACK TOGETHER…

Finally, the protective cover, Motor Shaft and final gear, and top plate!



28BYJ-48 MOTOR ELECTROMAGNETICS

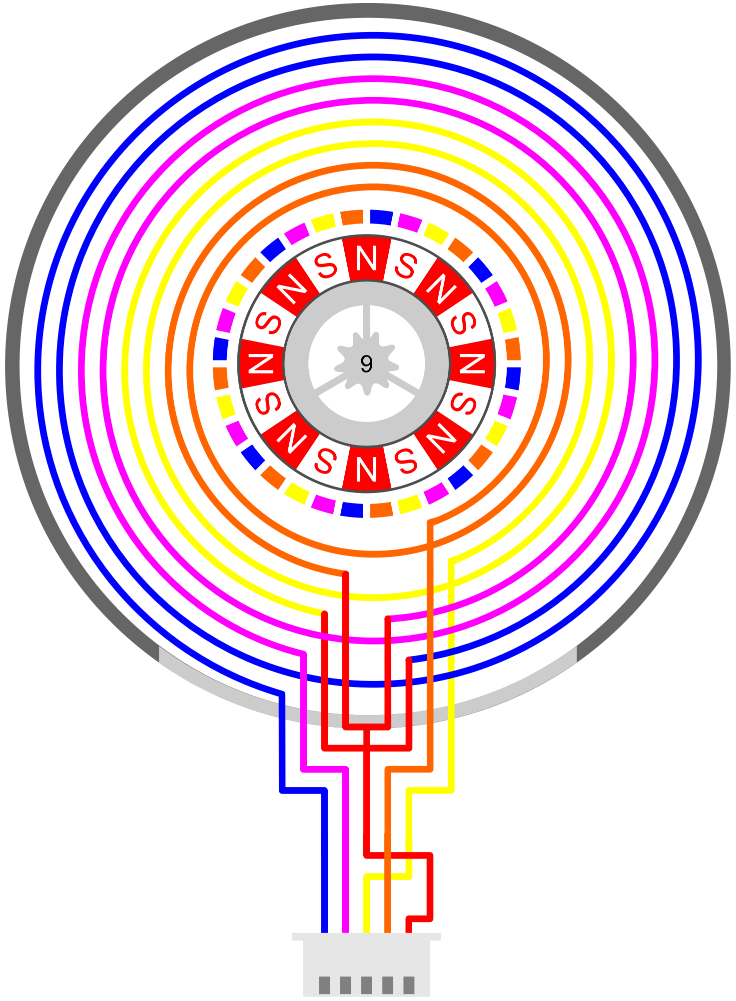


The 28BYJ-48 is a Unipolar Stepper Motor (as opposed to Bipolar).

With Unipolar motors, you don't need to change the polarity of the voltages on any give lead. That means that you don't need an H-Bridge to reverse the polarity.

The common center taps are connected to +5VDC. You then use some circuitry (the ULN2003 Array in this case) to pull the appropriate coil ends to ground to energize their respective half of the coil.

28BYJ-48 MOTOR ELECTROMAGNETICS



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With Unipolar motors, you don’t need to change the polarity of the voltages on any give lead. That means that you don’t need an H-Bridge to reverse the polarity.

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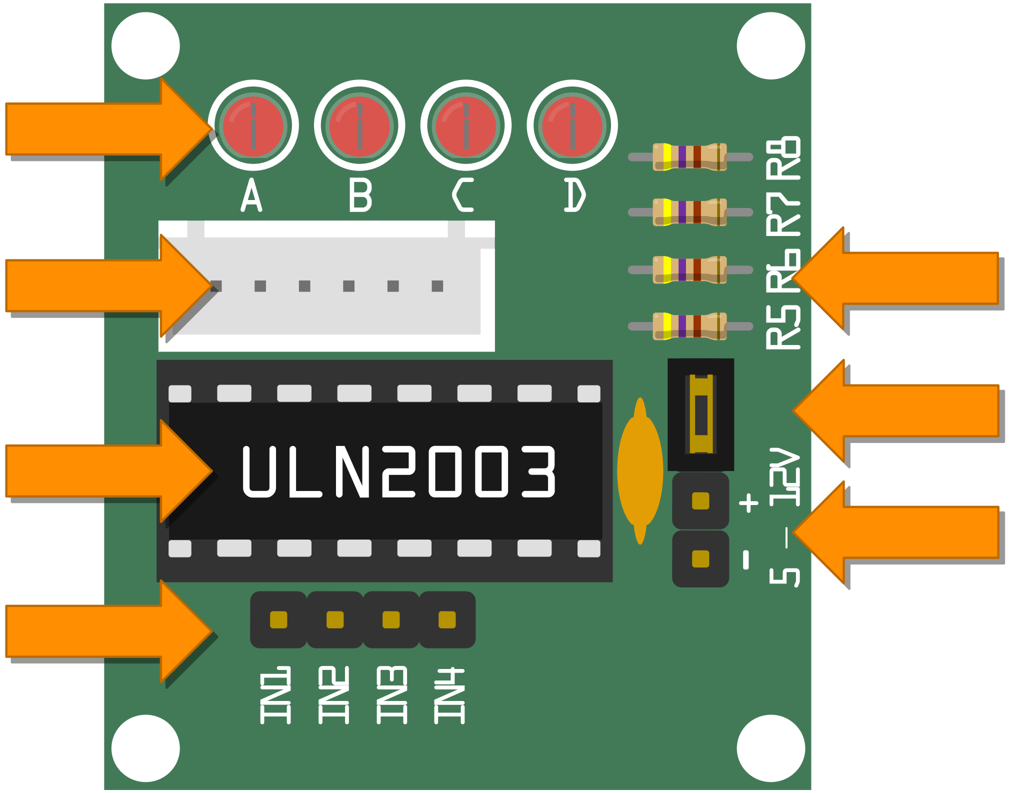
ULN2003 DARLINGTON ARRAY DRIVER



The ULN2003 houses an array of 7 Darlington Pairs

A Darlington Pair is basically a pair of transistors where the second transistor amplifies the output current of the first transistor.

This gives you a higher current gain than a single transistor and allows the low voltage, low current output of a microcontroller to drive the higher current stepper motor.

The board includes convenient IO and power headers, as well as coil power indicator LEDs

The ULN2003 actually supports seven input/output pairs, but the 28BYJ-48 only needs four (four phases) so only four pins are exposed.

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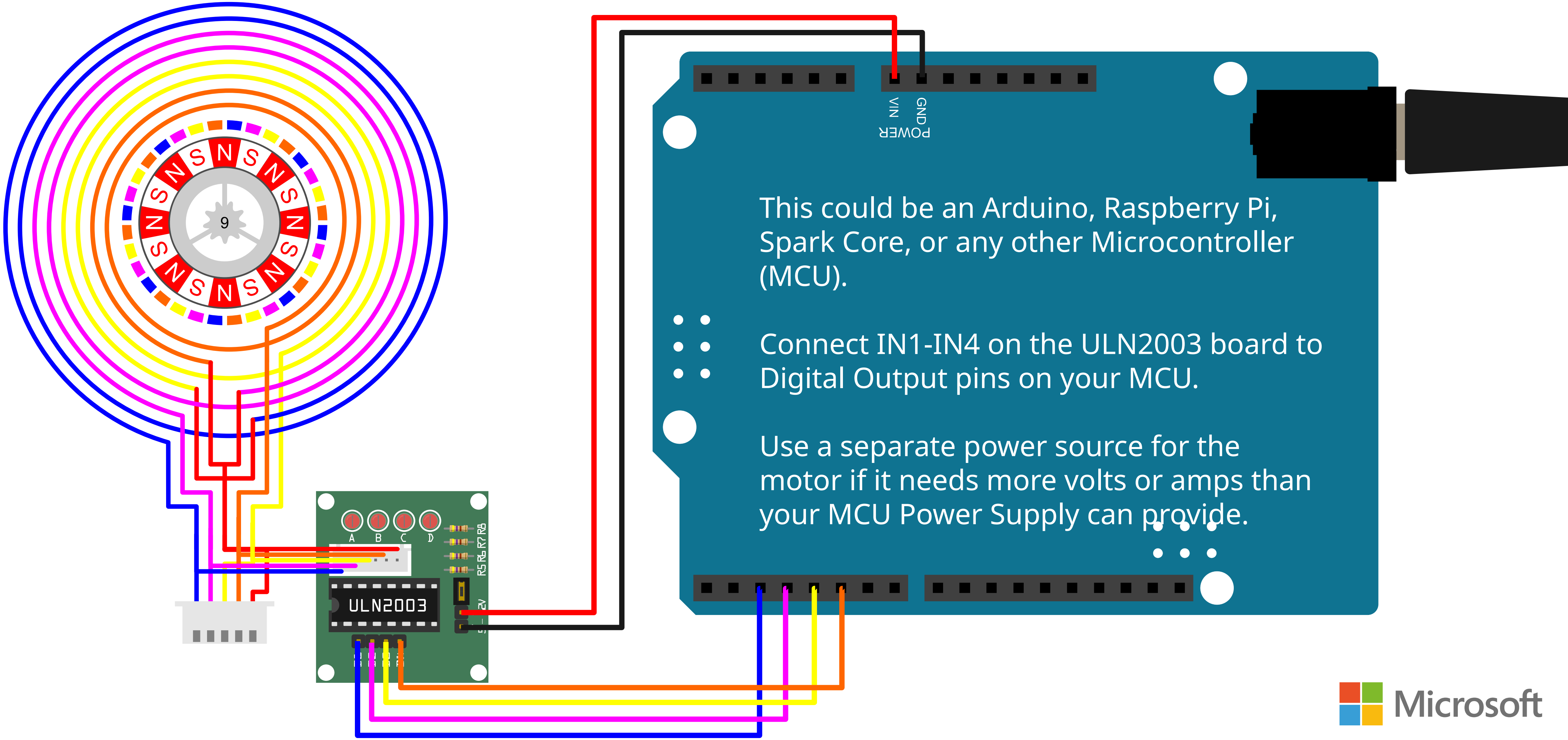
CONNECTING IT ALL UP…

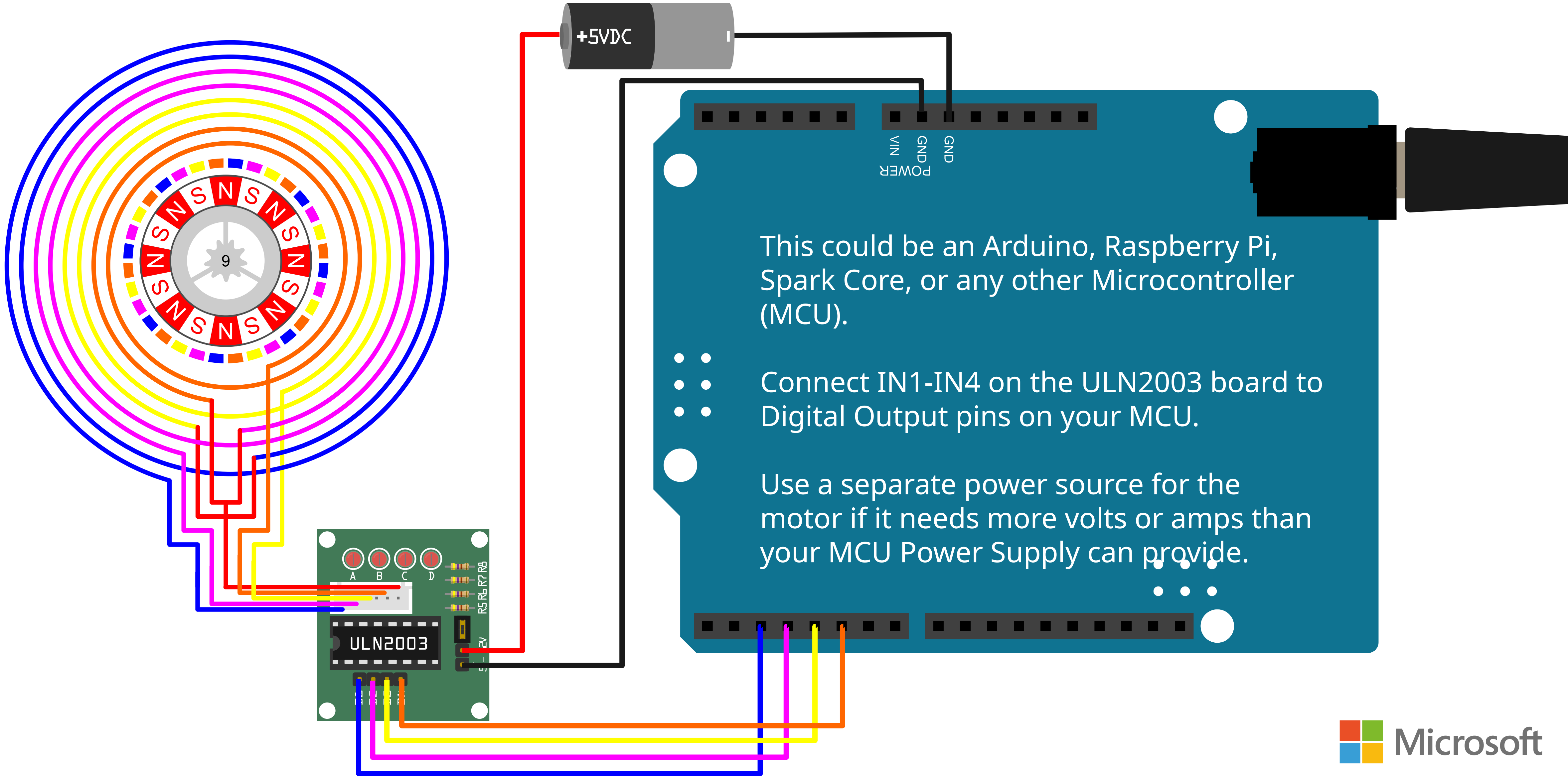


This could be an Arduino, Raspberry Pi, Spark Core, or any other Microcontroller (MCU).

Connect IN1-IN4 on the ULN2003 board to Digital Output pins on your MCU.

Use a separate power source for the motor if it needs more volts or amps than your MCU Power Supply can provide.

USING A SEPARATE POWER SOURCE…



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Use a separate power source for the motor if it needs more volts or amps than your MCU Power Supply can provide.

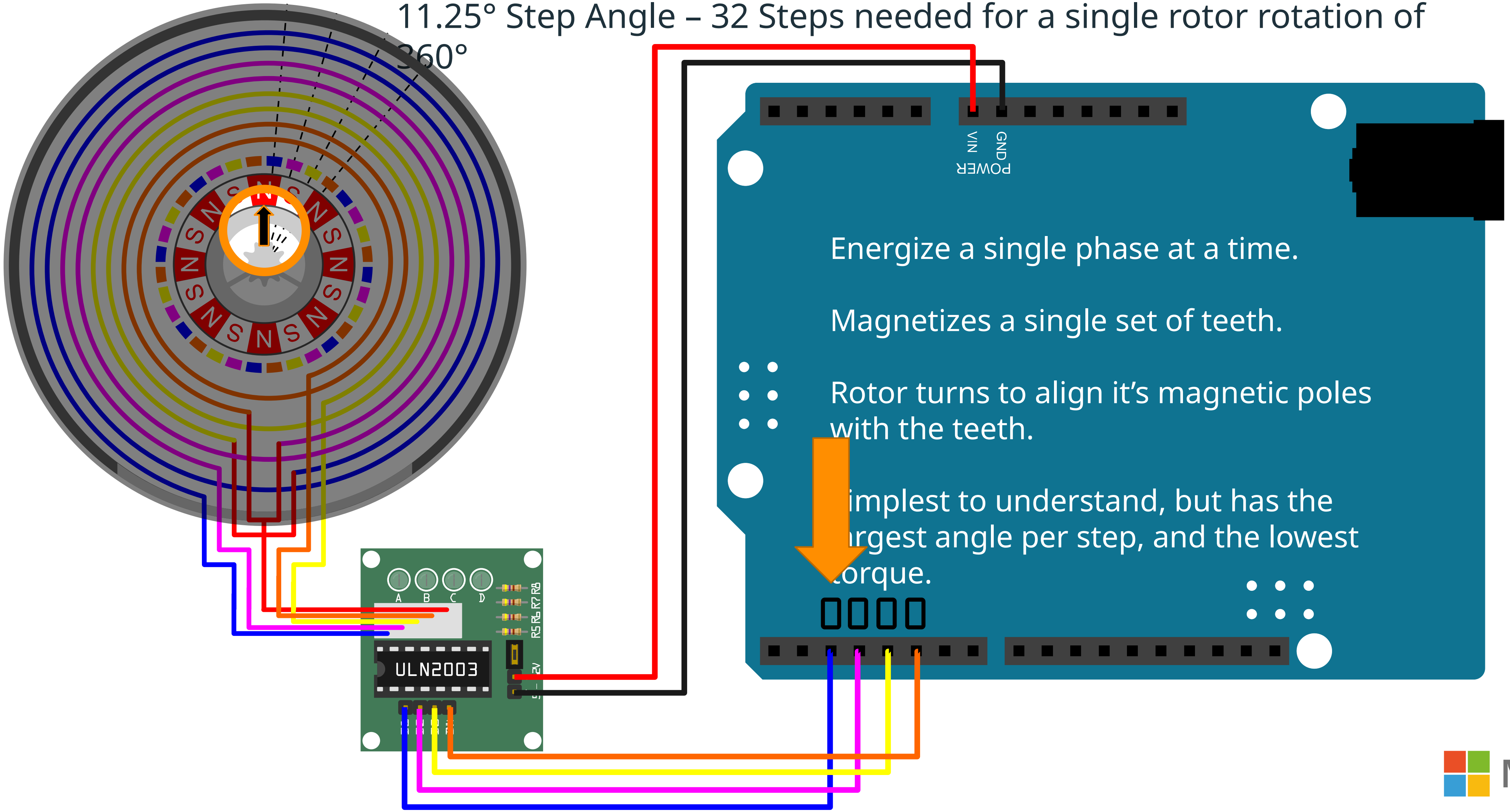
DRIVING METHODS

STEPPING METHODS

		STEPS							
METHODS	PHASES	1	2	3	4	5	6	7	8
WAVE DRIVE One phase at a time Simplest, but least used	BLUE	1				1			
	PINK		1				1		
	YELLOW			1				1	
	ORANGE				1				1
FULL STEP Two phases at a time Strongest Torque	BLUE	1			1	1			1
	PINK	1	1			1	1		
	YELLOW		1	1			1	1	
	ORANGE			1	1			1	1
HALF STEP One, or two phases at a time Smallest step angle Medium torque	BLUE	1	1						1
	PINK		1	1	1				
	YELLOW				1	1	1		
	ORANGE						1	1	1

WAVE DRIVING…

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



Energize a single phase at a time.

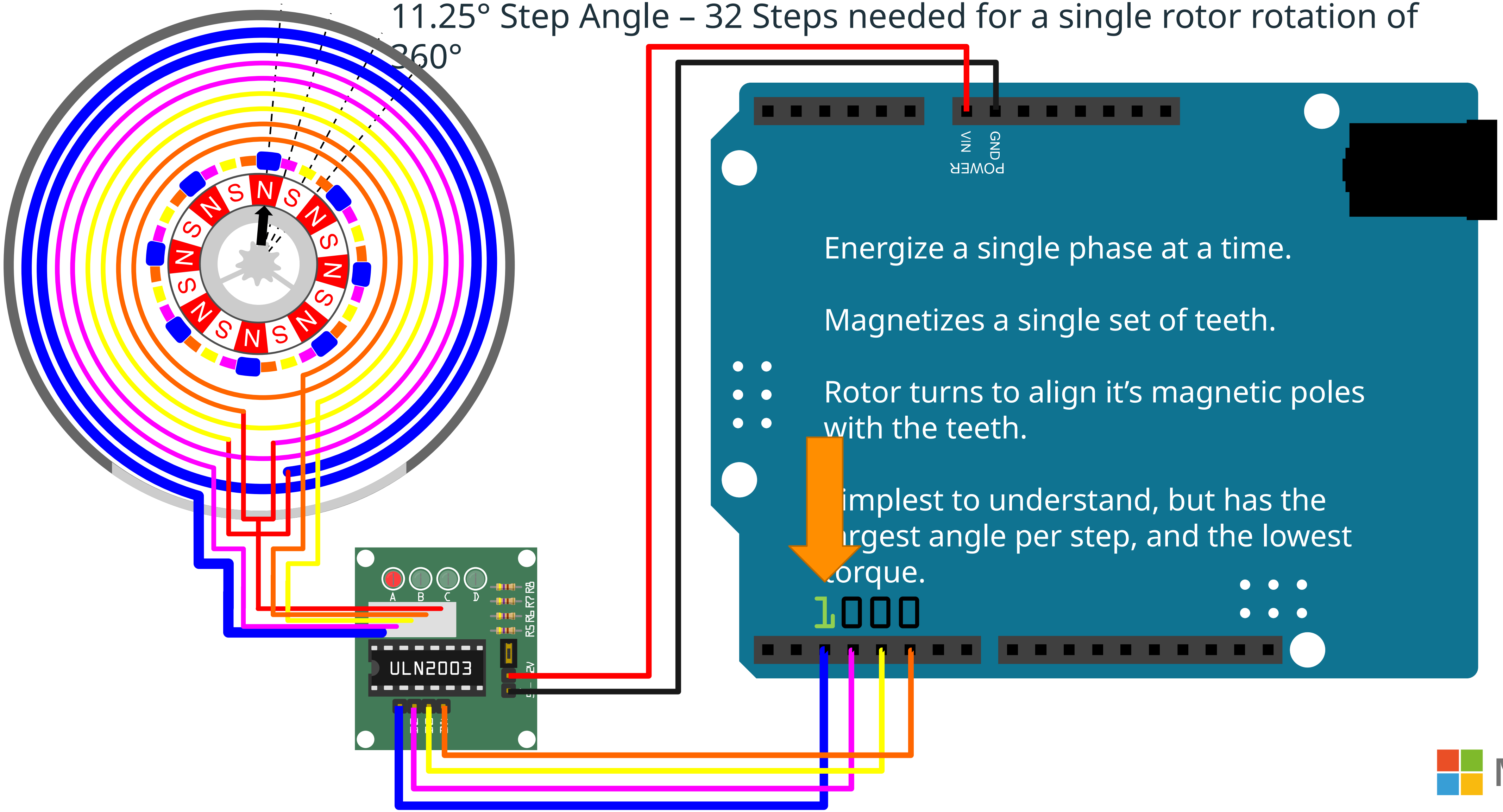
Magnetizes a single set of teeth.

Rotor turns to align it's magnetic poles with the teeth.

Simplest to understand, but has the largest angle per step, and the lowest torque.

WAVE DRIVING – STEP 1

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



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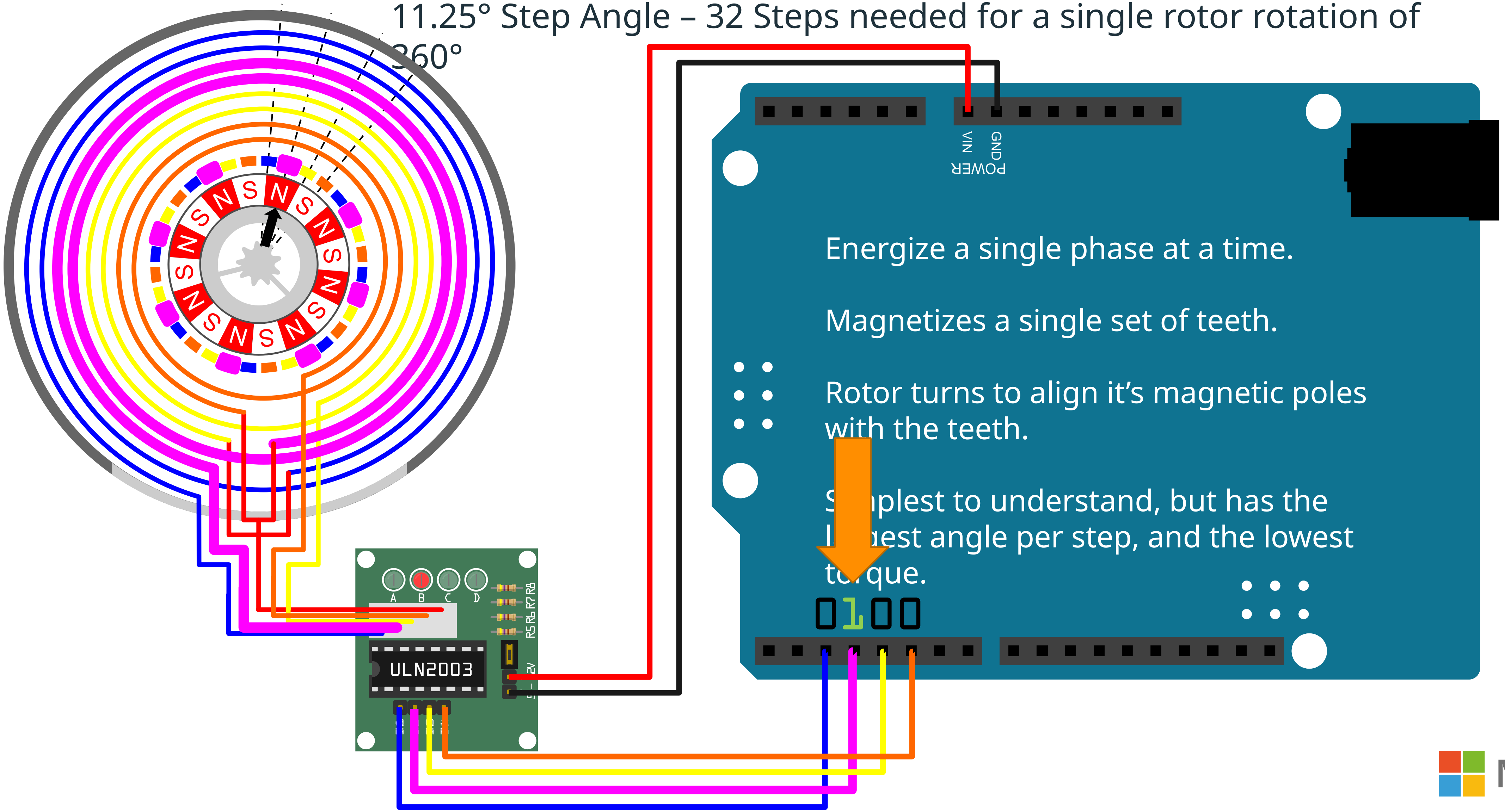
WAVE DRIVING – STEP 2

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



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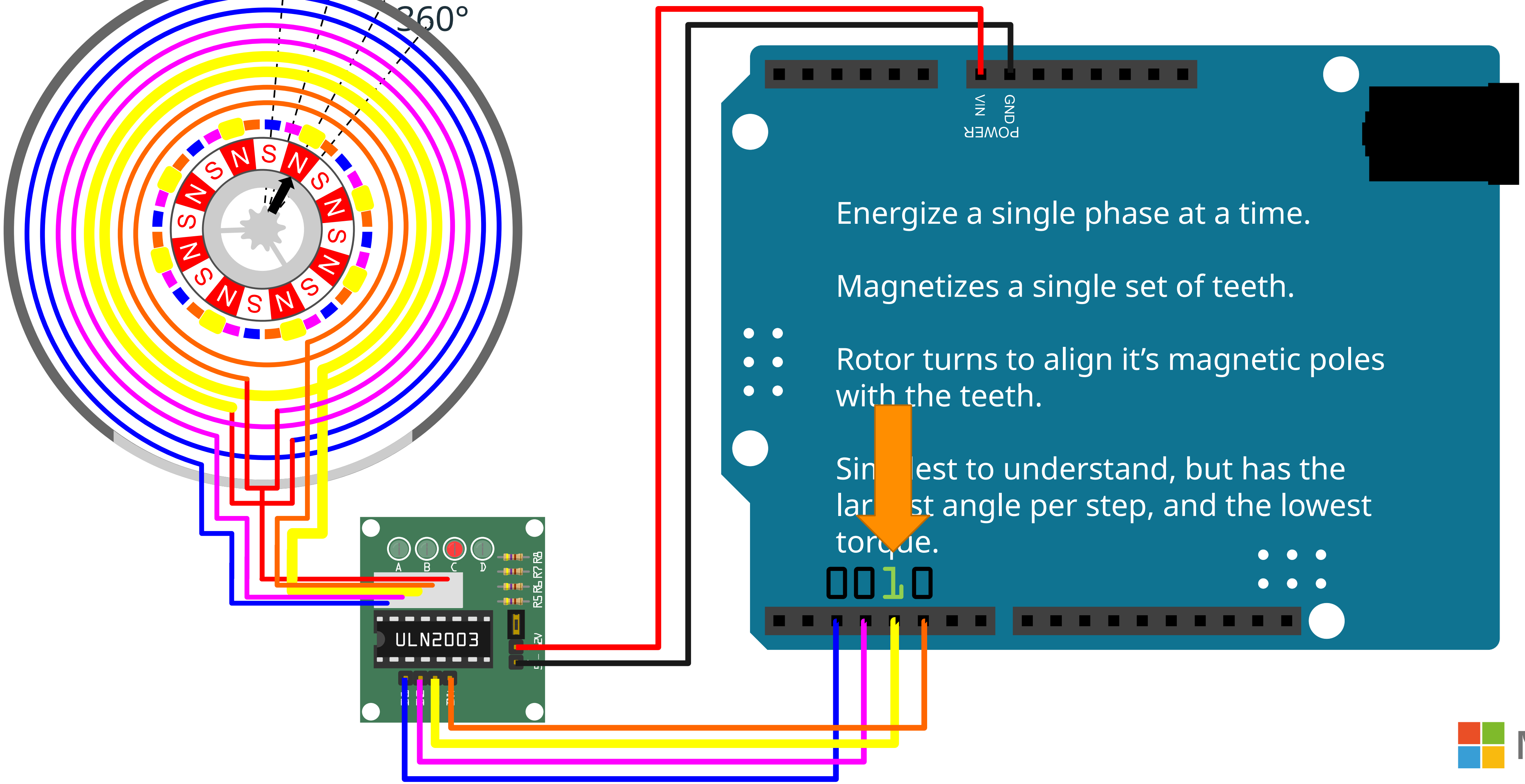
Simplest to understand, but has the largest angle per step, and the lowest torque.

WAVE DRIVING – STEP 3

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



Energize a single phase at a time.

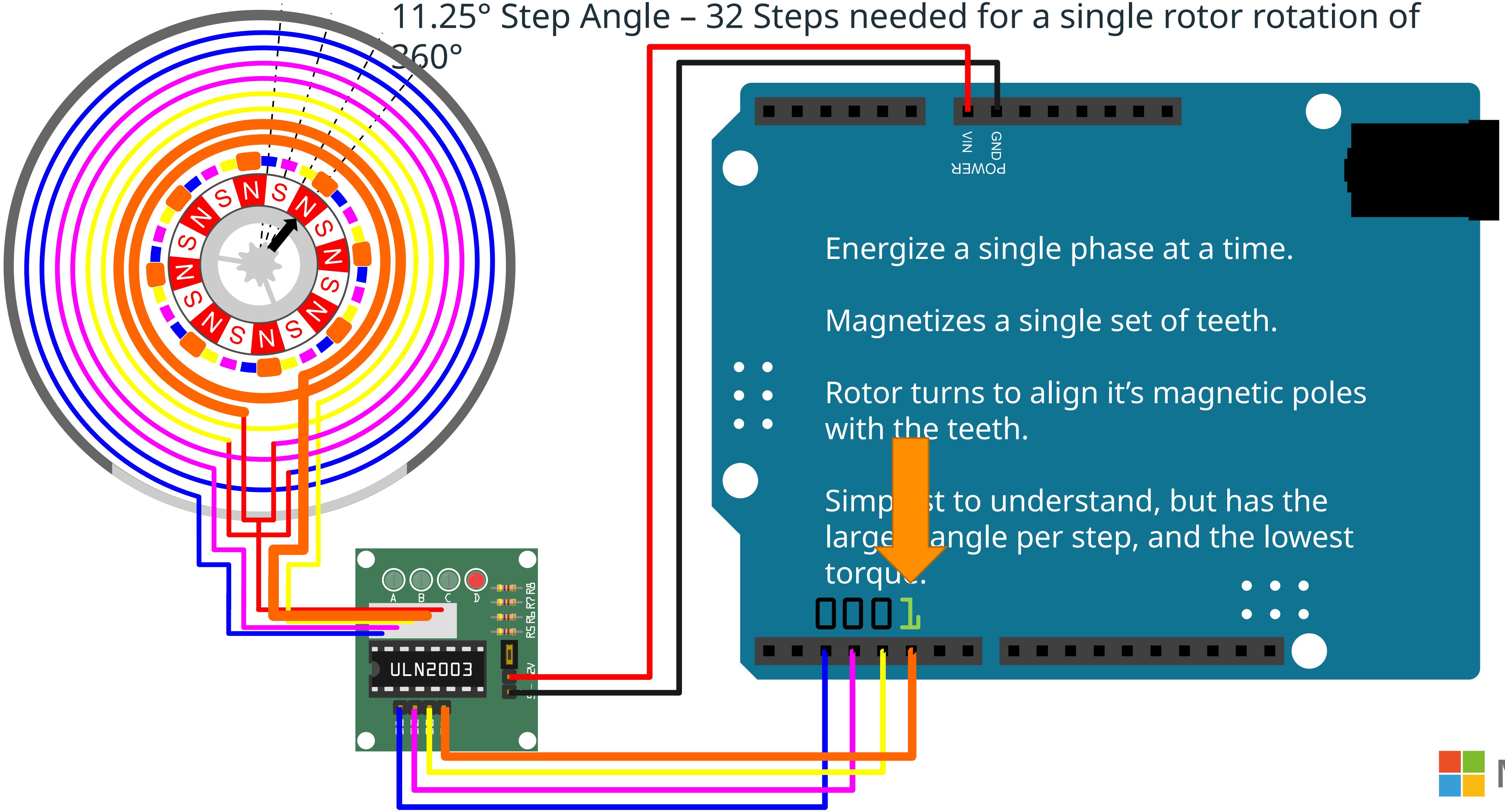
Magnetizes a single set of teeth.

Rotor turns to align it’s magnetic poles with the teeth.

Simplest to understand, but has the largest angle per step, and the lowest torque.

WAVE DRIVING – STEP 4

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



Energize a single phase at a time.

Magnetizes a single set of teeth.

Rotor turns to align it’s magnetic poles with the teeth.

Simplest to understand, but has the largest angle per step, and the lowest torque.

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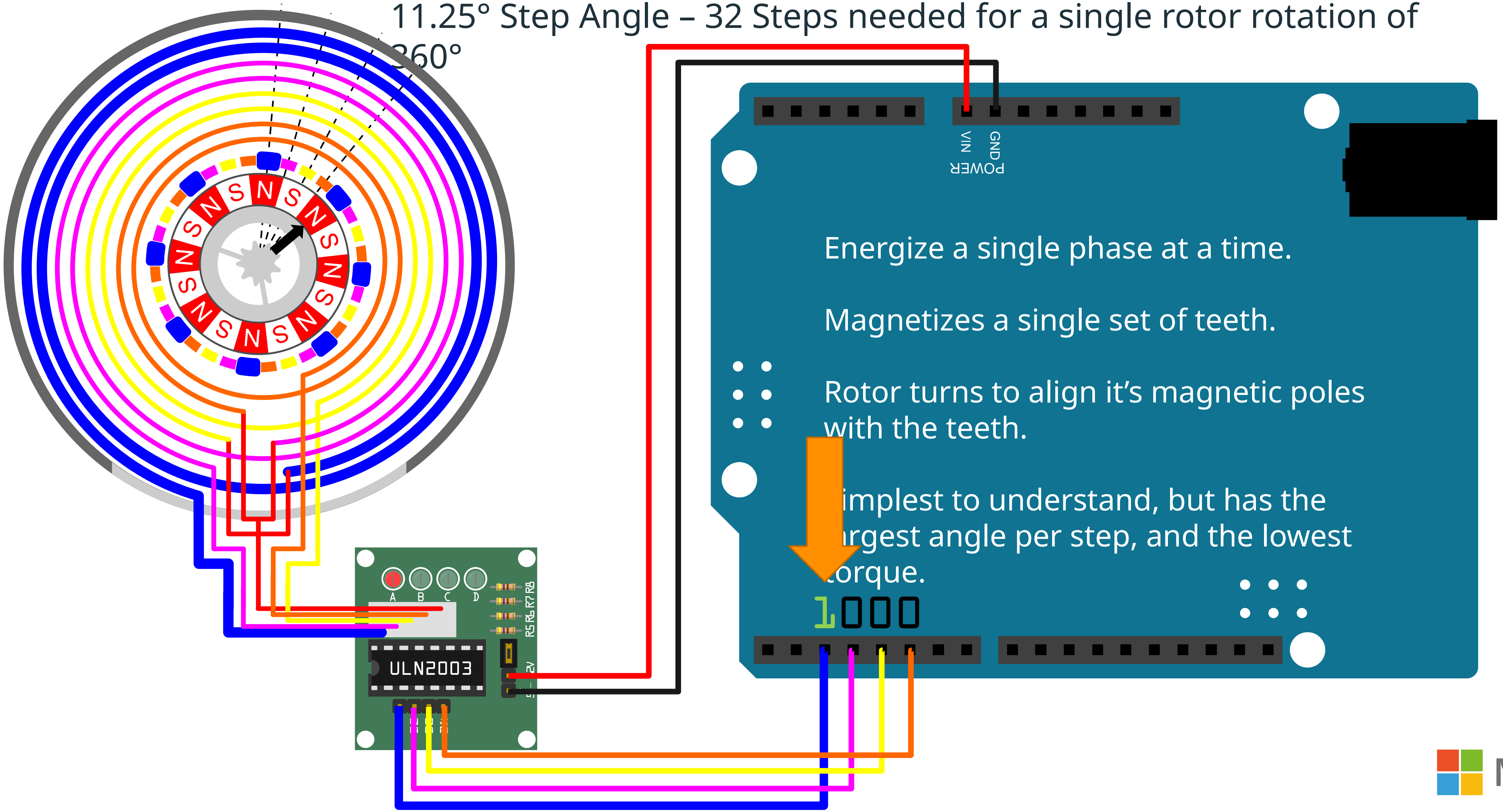
WAVE DRIVING – STEP 5

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



Energize a single phase at a time.

Magnetizes a single set of teeth.

Rotor turns to align it’s magnetic poles with the teeth.

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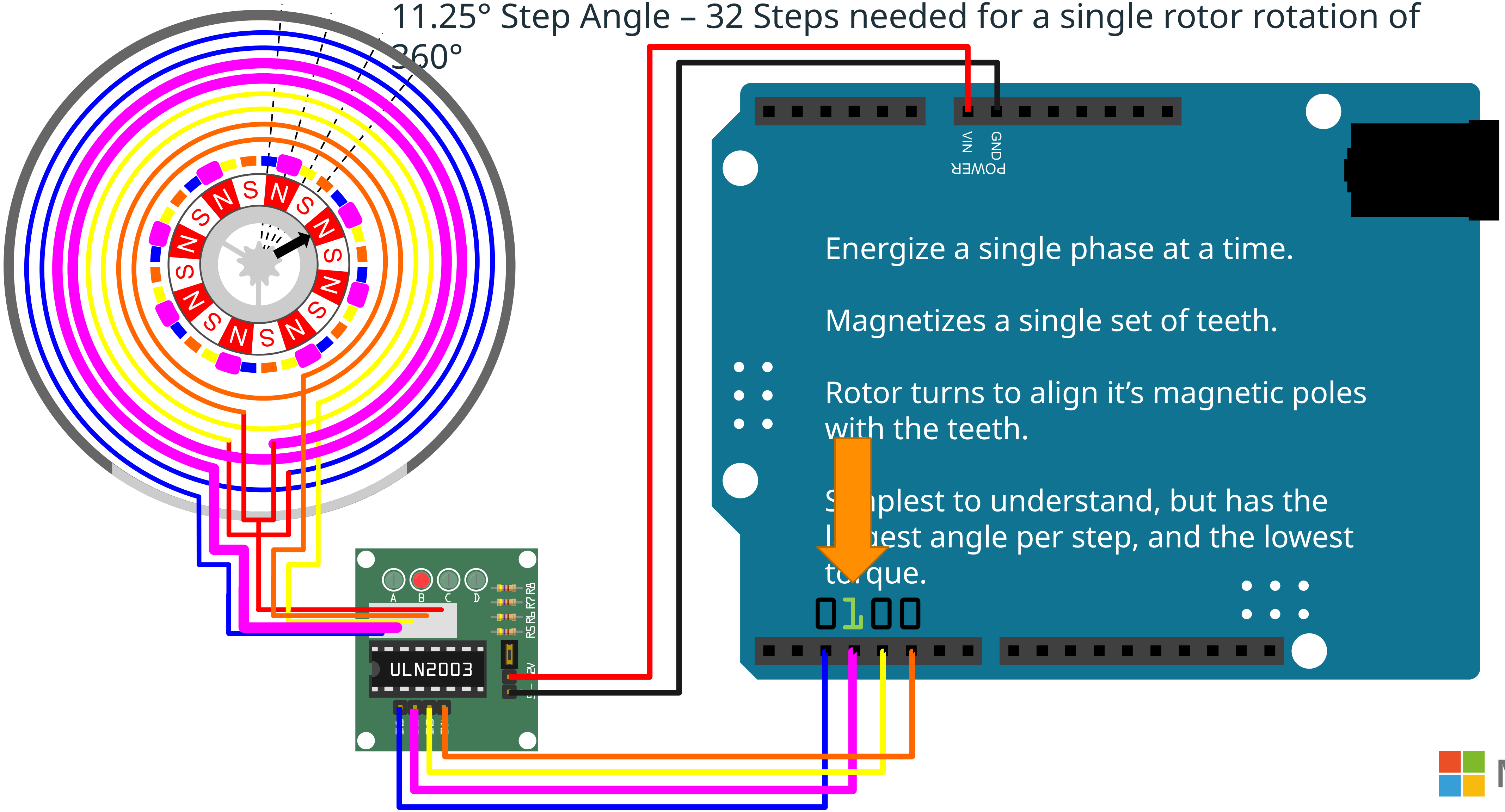
WAVE DRIVING – STEP 6

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



Energize a single phase at a time.

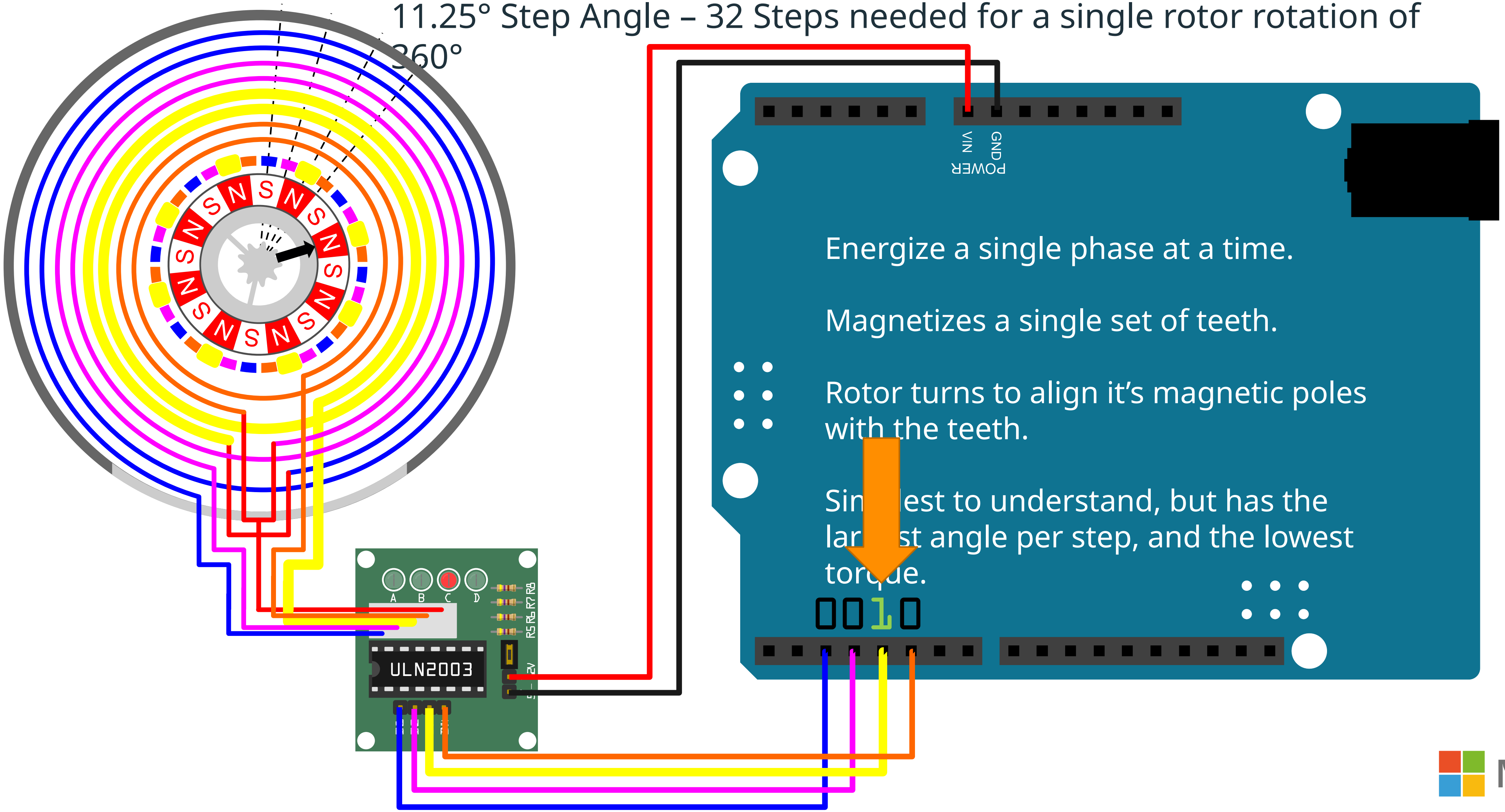
Magnetizes a single set of teeth.

Rotor turns to align it’s magnetic poles with the teeth.

Simplest to understand, but has the largest angle per step, and the lowest torque.

WAVE DRIVING – STEP 7

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



Energize a single phase at a time.

Magnetizes a single set of teeth.

Rotor turns to align it’s magnetic poles with the teeth.

Simplest to understand, but has the largest angle per step, and the lowest torque.

WAVE DRIVING – STEP 8

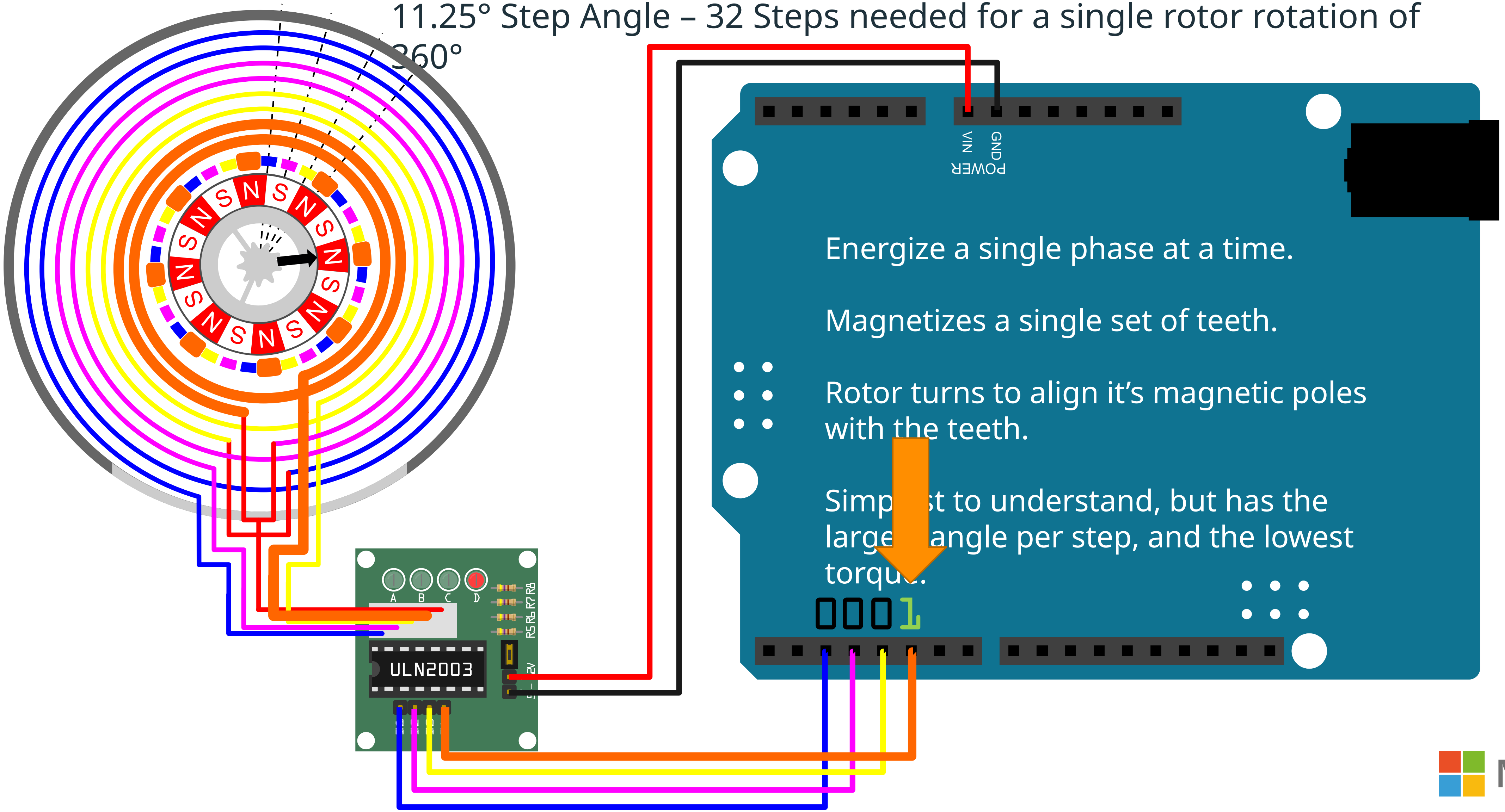
11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°

Energize a single phase at a time.

Magnetizes a single set of teeth.

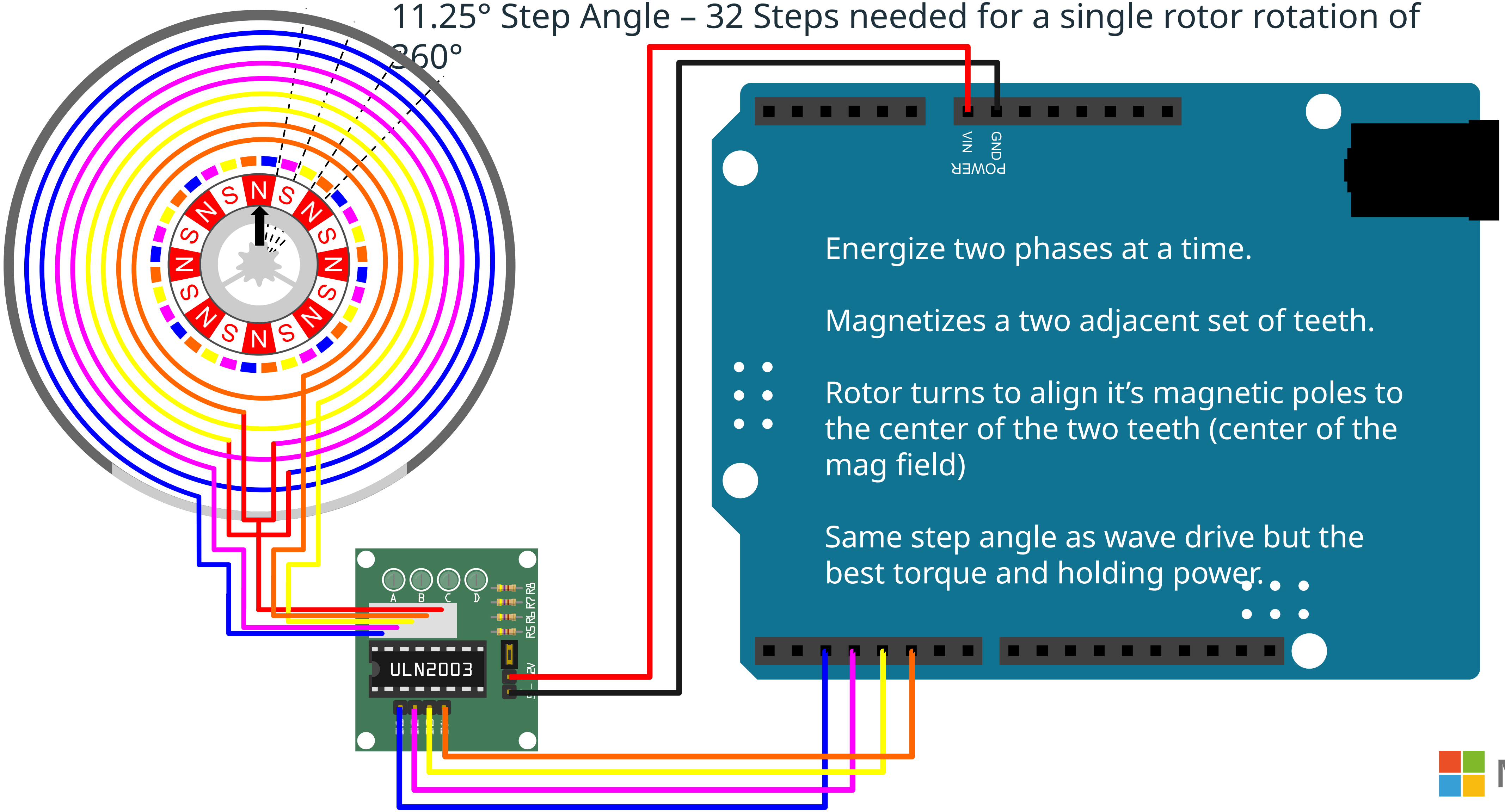
Rotor turns to align it’s magnetic poles with the teeth.

Simplest to understand, but has the largest angle per step, and the lowest torque.



FULL STEPPING…

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



Energize two phases at a time.

Magnetizes a two adjacent set of teeth.

Rotor turns to align it’s magnetic poles to the center of the two teeth (center of the mag field)

Same step angle as wave drive but the best torque and holding power.

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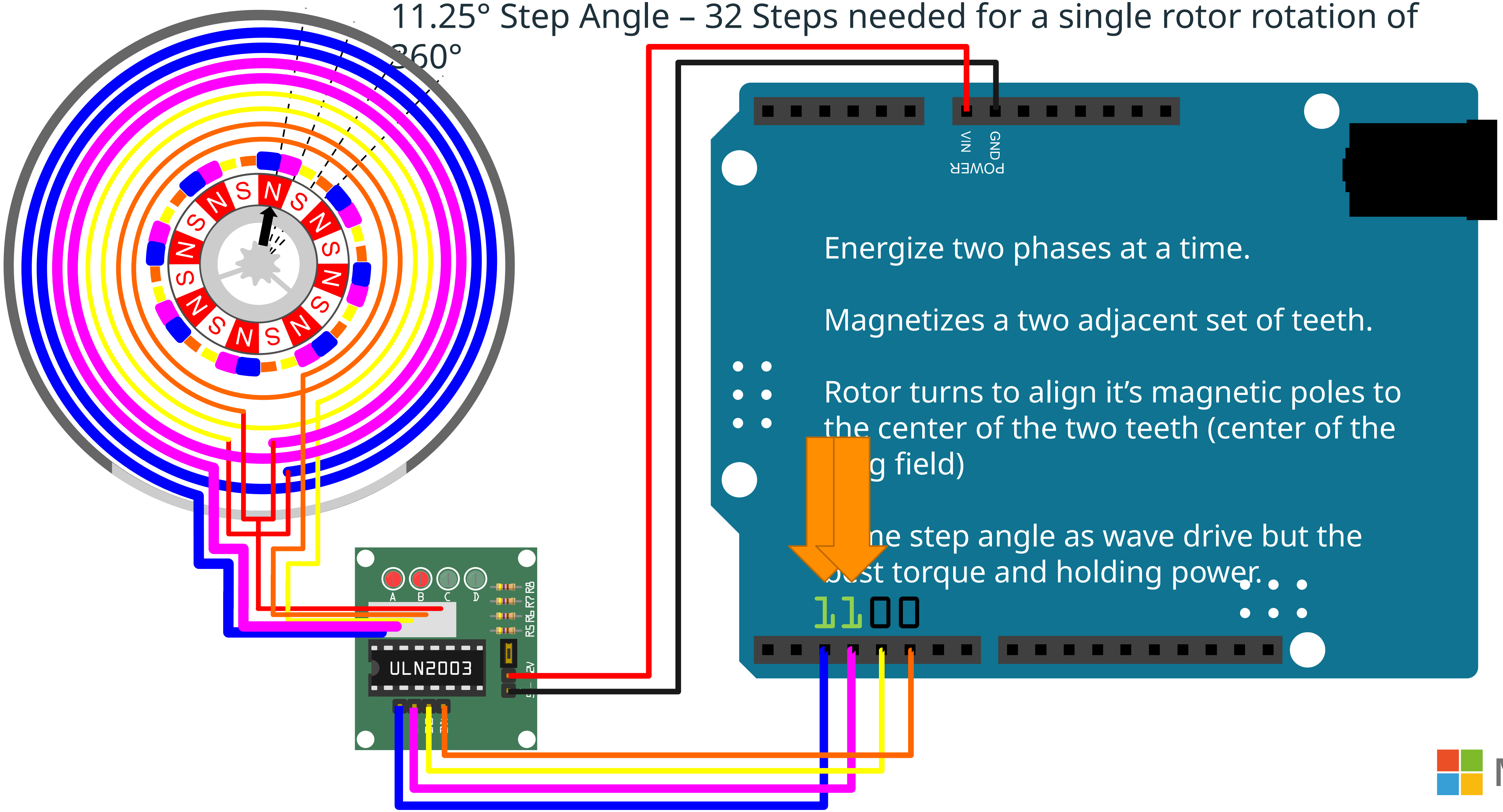
FULL STEPPING – STEP 1

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



Energize two phases at a time.

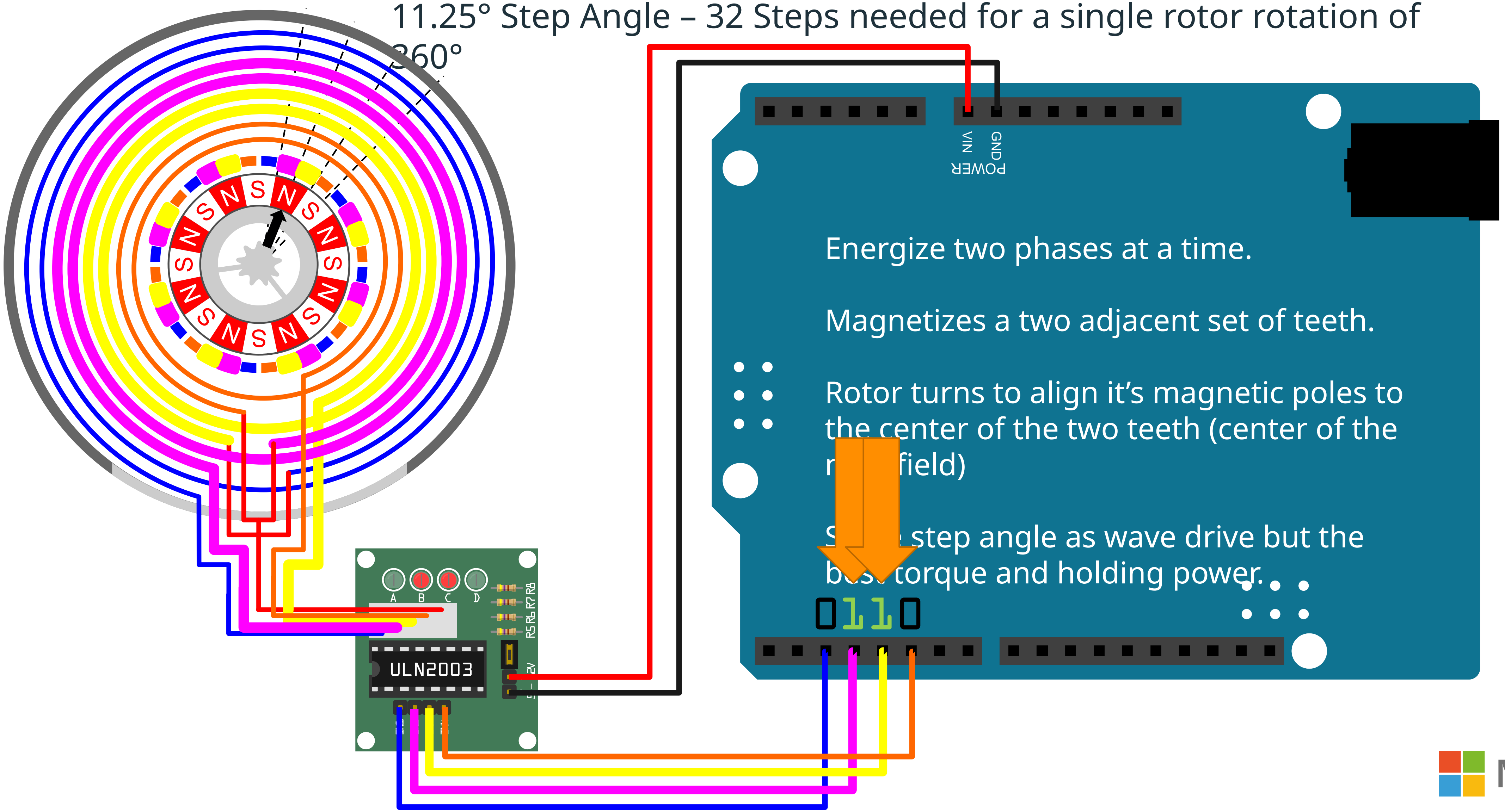
Magnetizes a two adjacent set of teeth.

Rotor turns to align it’s magnetic poles to the center of the two teeth (center of the …g field)

…ne step angle as wave drive but the …st torque and holding power.

FULL STEPPING – STEP 2

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



Energize two phases at a time.

Magnetizes a two adjacent set of teeth.

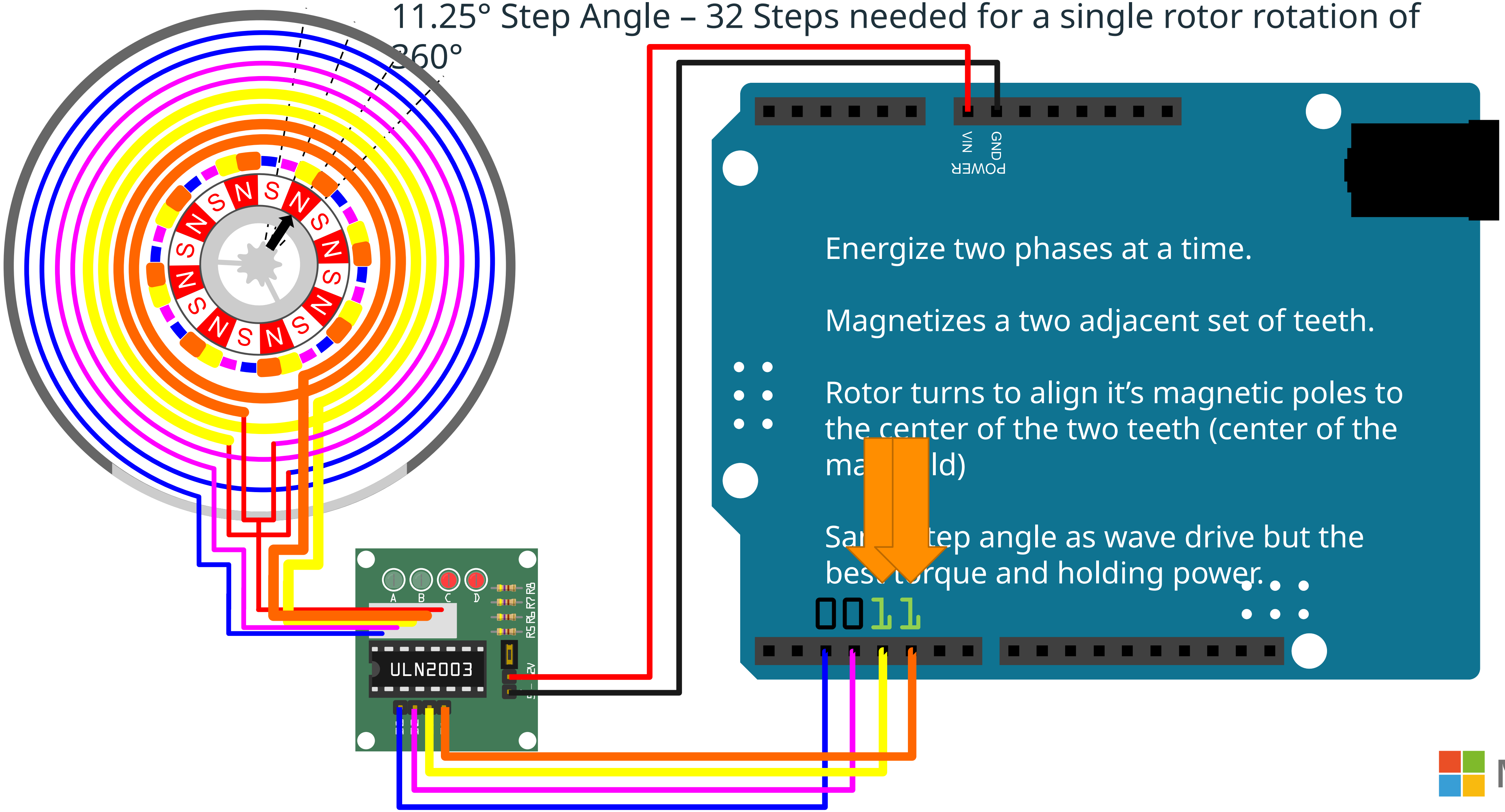
Rotor turns to align it’s magnetic poles to the center of the two teeth (center of the r[illegible] field)

S[illegible] step angle as wave drive but the b[illegible] torque and holding power.

Microsoft

FULL STEPPING – STEP 3

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



Energize two phases at a time.

Magnetizes a two adjacent set of teeth.

Rotor turns to align it's magnetic poles to the center of the two teeth (center of the ma ld)

Sa tep angle as wave drive but the bes orque and holding power.

FULL STEPPING – STEP 4

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°

Energize two phases at a time.

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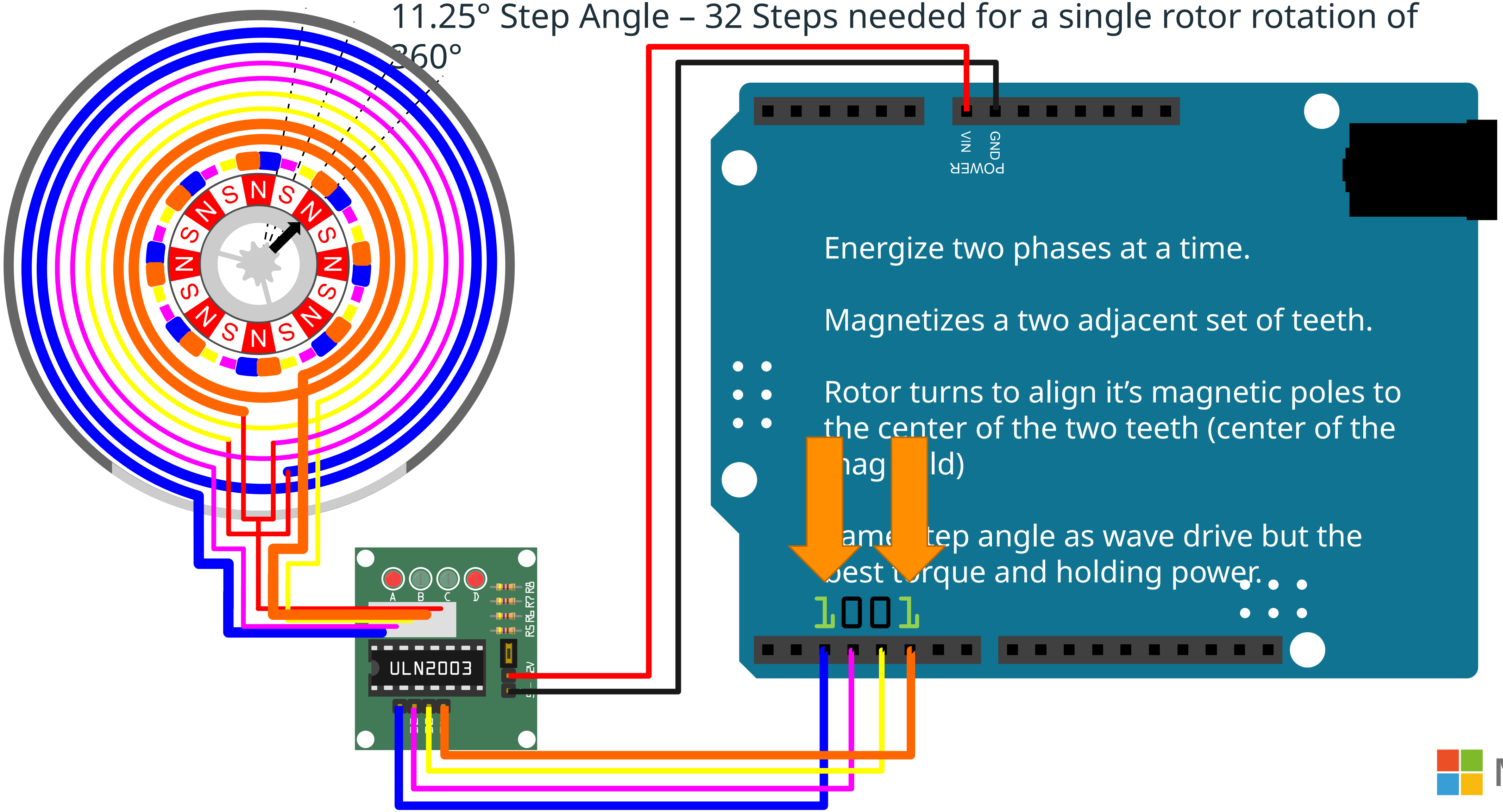
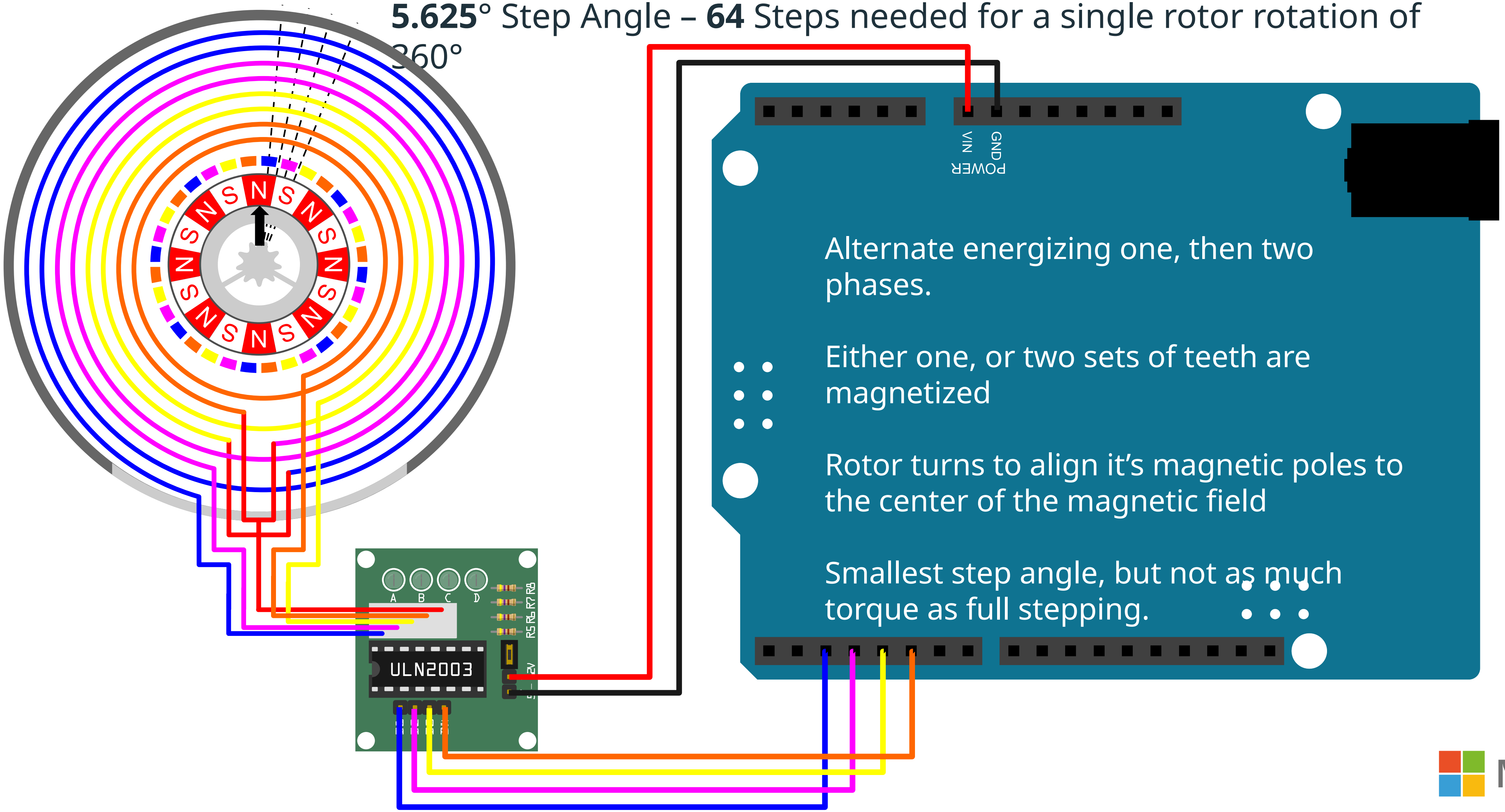
Rotor turns to align it's magnetic poles to the center of the two teeth (center of the mag ld)

ame tep angle as wave drive but the est t rque and holding power.



HALF STEPPING…

5.625° Step Angle – **64** Steps needed for a single rotor rotation of 360°



Alternate energizing one, then two phases.

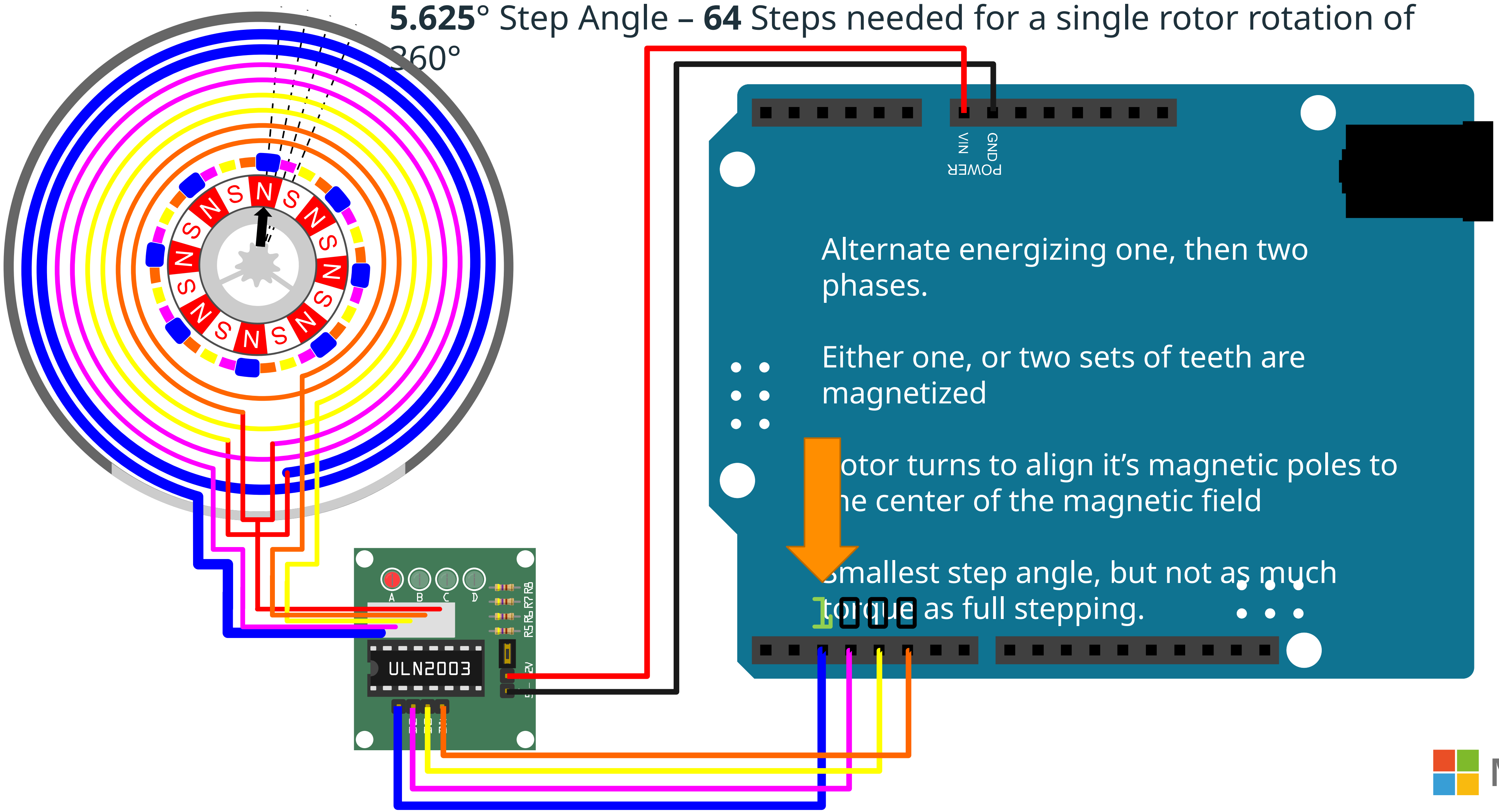
Either one, or two sets of teeth are magnetized

Rotor turns to align it's magnetic poles to the center of the magnetic field

Smallest step angle, but not as much torque as full stepping.

HALF STEPPING – STEP 1

5.625° Step Angle – **64** Steps needed for a single rotor rotation of 360°



Alternate energizing one, then two phases.

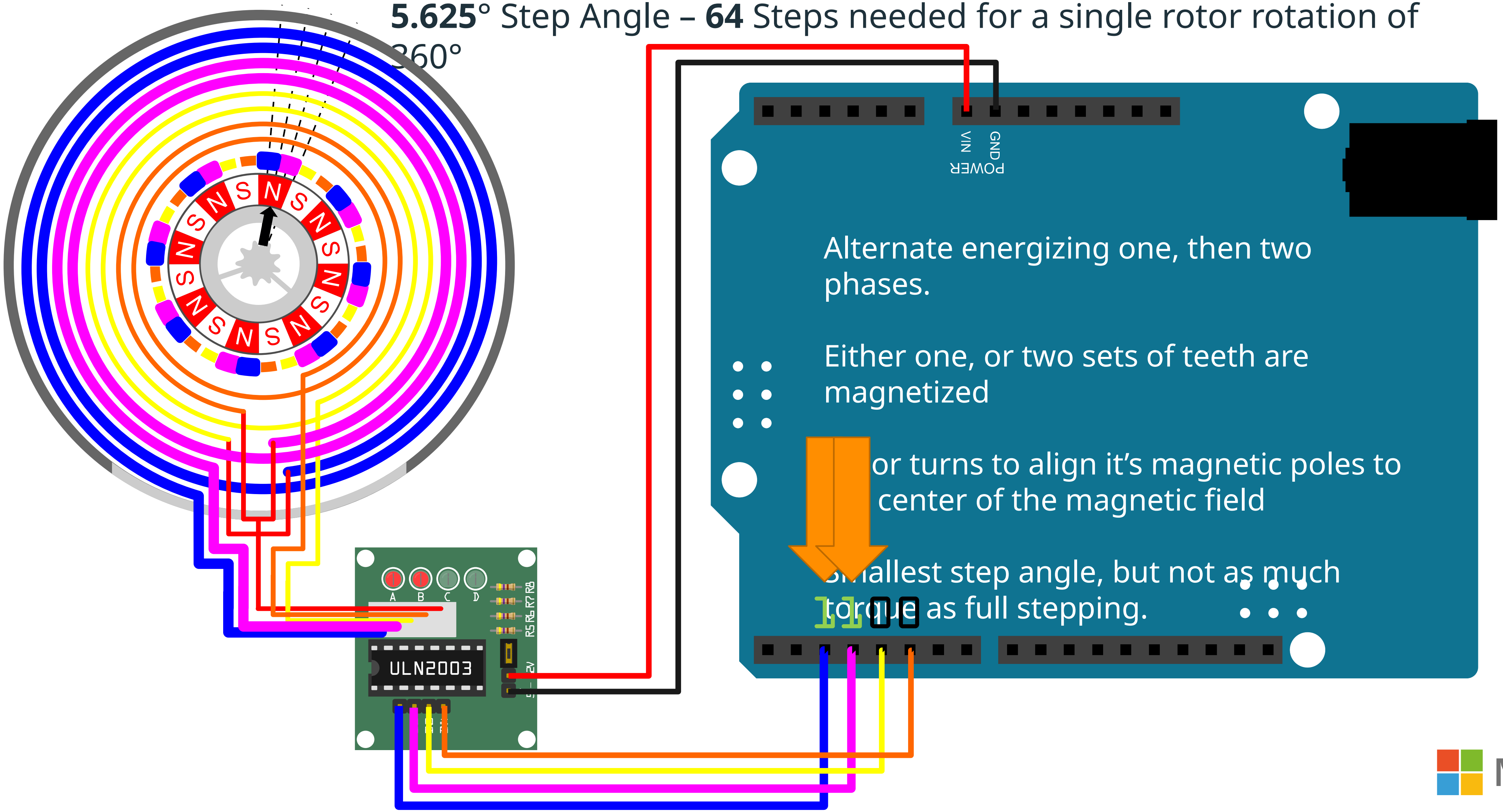
Either one, or two sets of teeth are magnetized

Rotor turns to align it's magnetic poles to the center of the magnetic field

Smallest step angle, but not as much torque as full stepping.

HALF STEPPING – STEP 2

5.625° Step Angle – **64** Steps needed for a single rotor rotation of 360°



Alternate energizing one, then two phases.

Either one, or two sets of teeth are magnetized

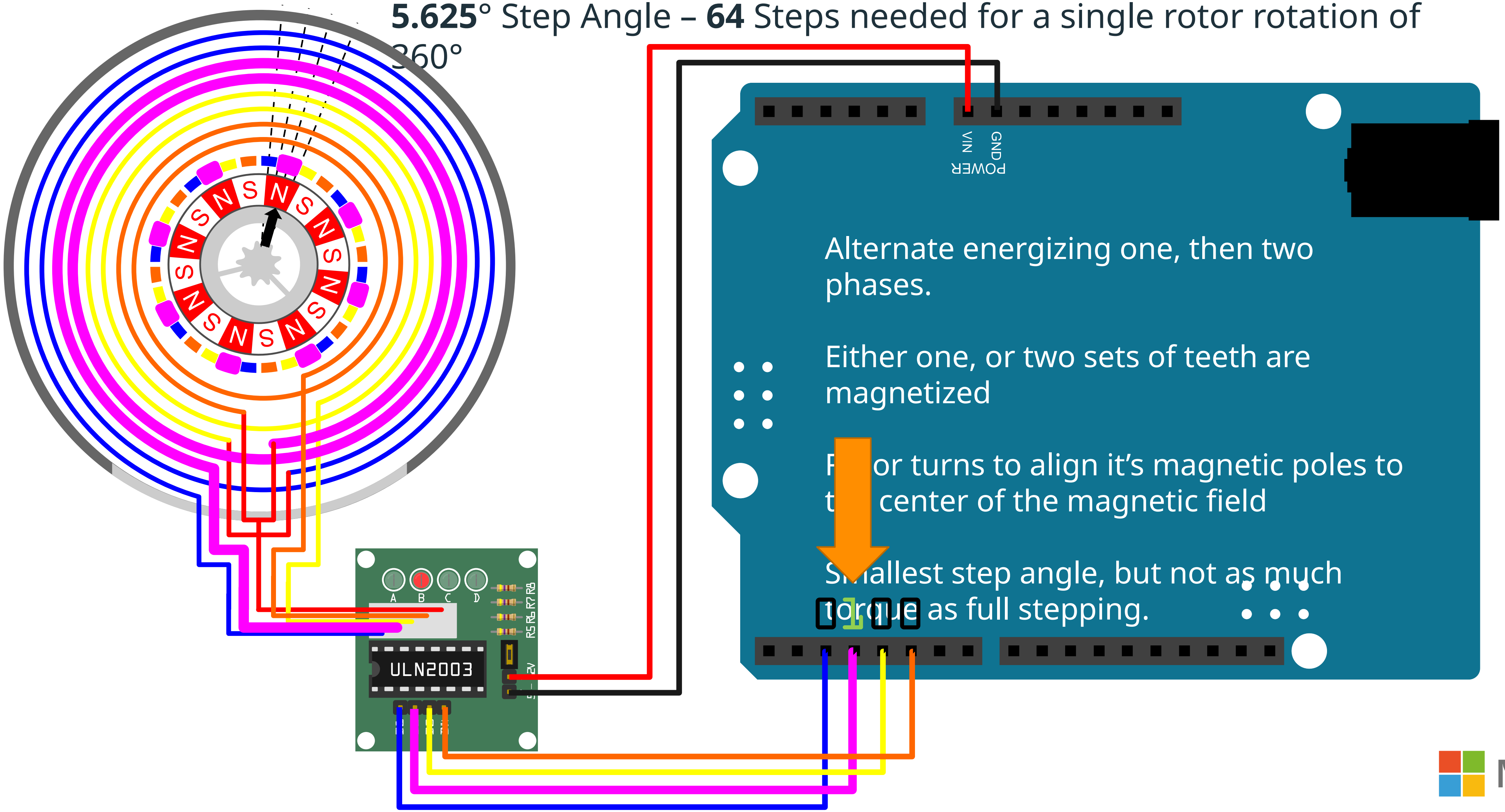
or turns to align it's magnetic poles to center of the magnetic field

Smallest step angle, but not as much torque as full stepping.

Microsoft

HALF STEPPING – STEP 3

5.625° Step Angle – **64** Steps needed for a single rotor rotation of 360°



Alternate energizing one, then two phases.

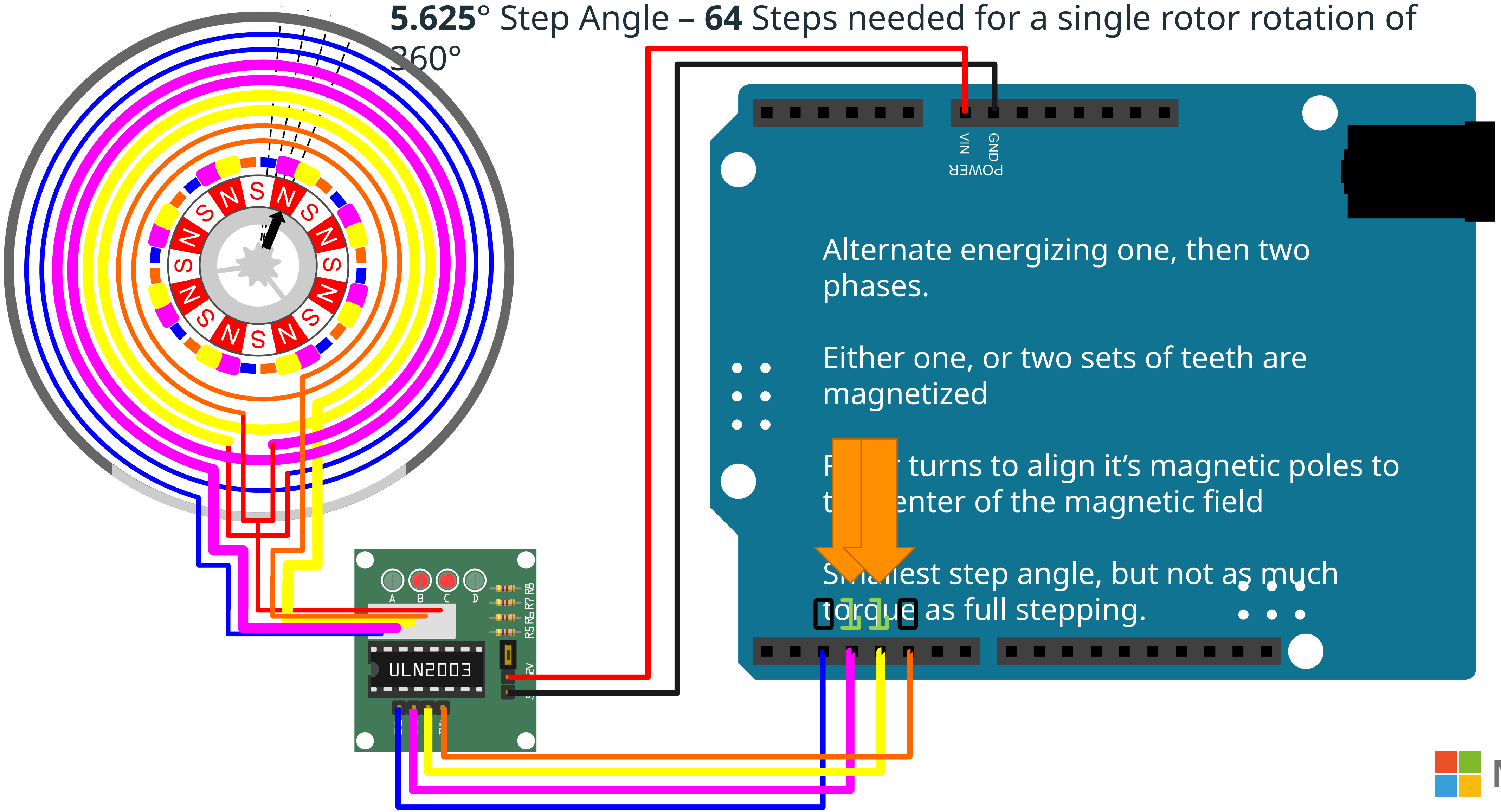
Either one, or two sets of teeth are magnetized

Rotor turns to align it's magnetic poles to the center of the magnetic field

Smallest step angle, but not as much torque as full stepping.

HALF STEPPING – STEP 4

5.625° Step Angle – **64** Steps needed for a single rotor rotation of 360°



Alternate energizing one, then two phases.

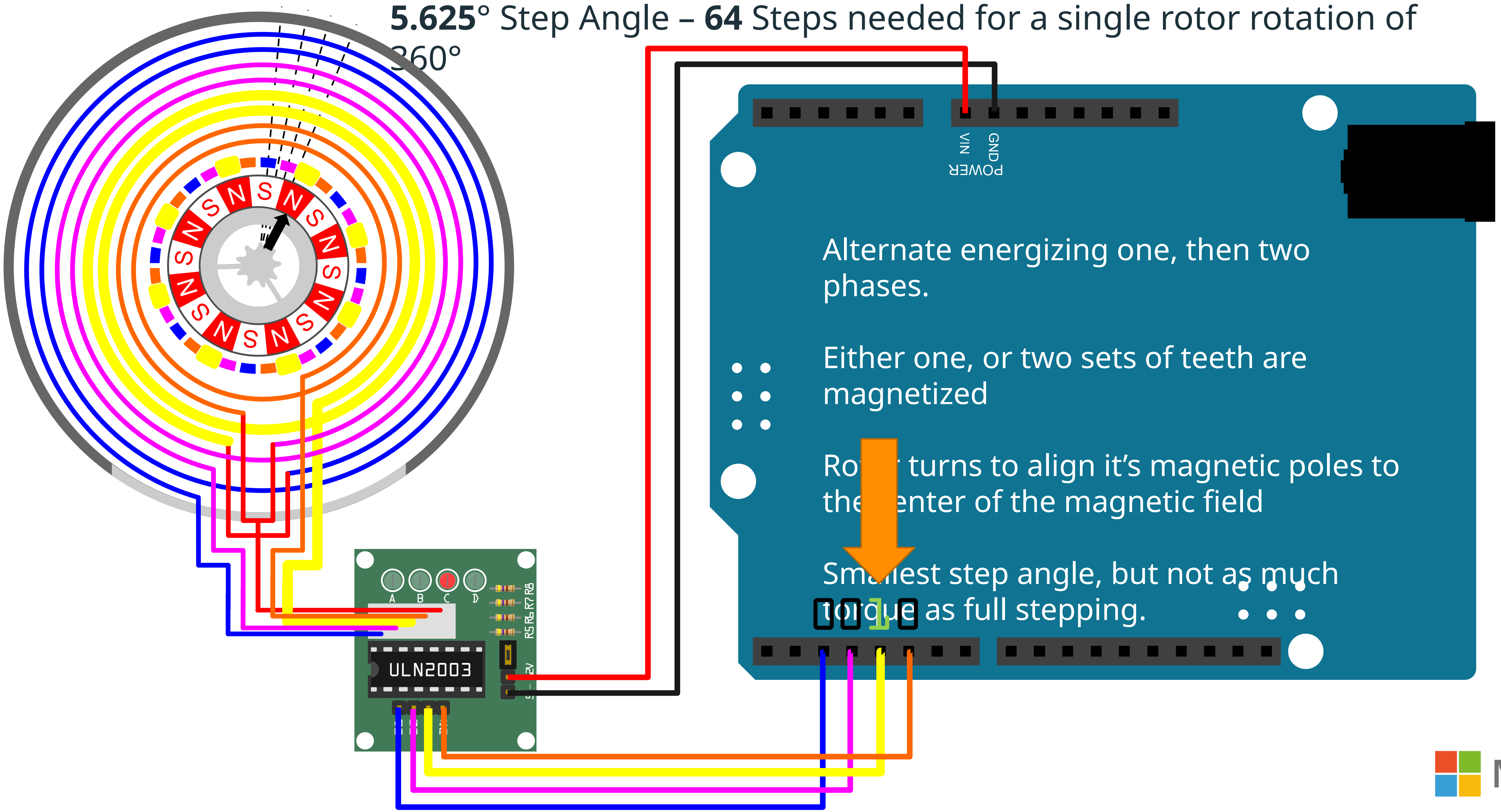
Either one, or two sets of teeth are magnetized

Rotor turns to align it's magnetic poles to the center of the magnetic field

Smallest step angle, but not as much torque as full stepping.

HALF STEPPING – STEP 5

5.625° Step Angle – **64** Steps needed for a single rotor rotation of 360°



Alternate energizing one, then two phases.

Either one, or two sets of teeth are magnetized

Rotor turns to align it's magnetic poles to the center of the magnetic field

Smallest step angle, but not as much torque as full stepping.

Microsoft

HALF STEPPING – STEP 6

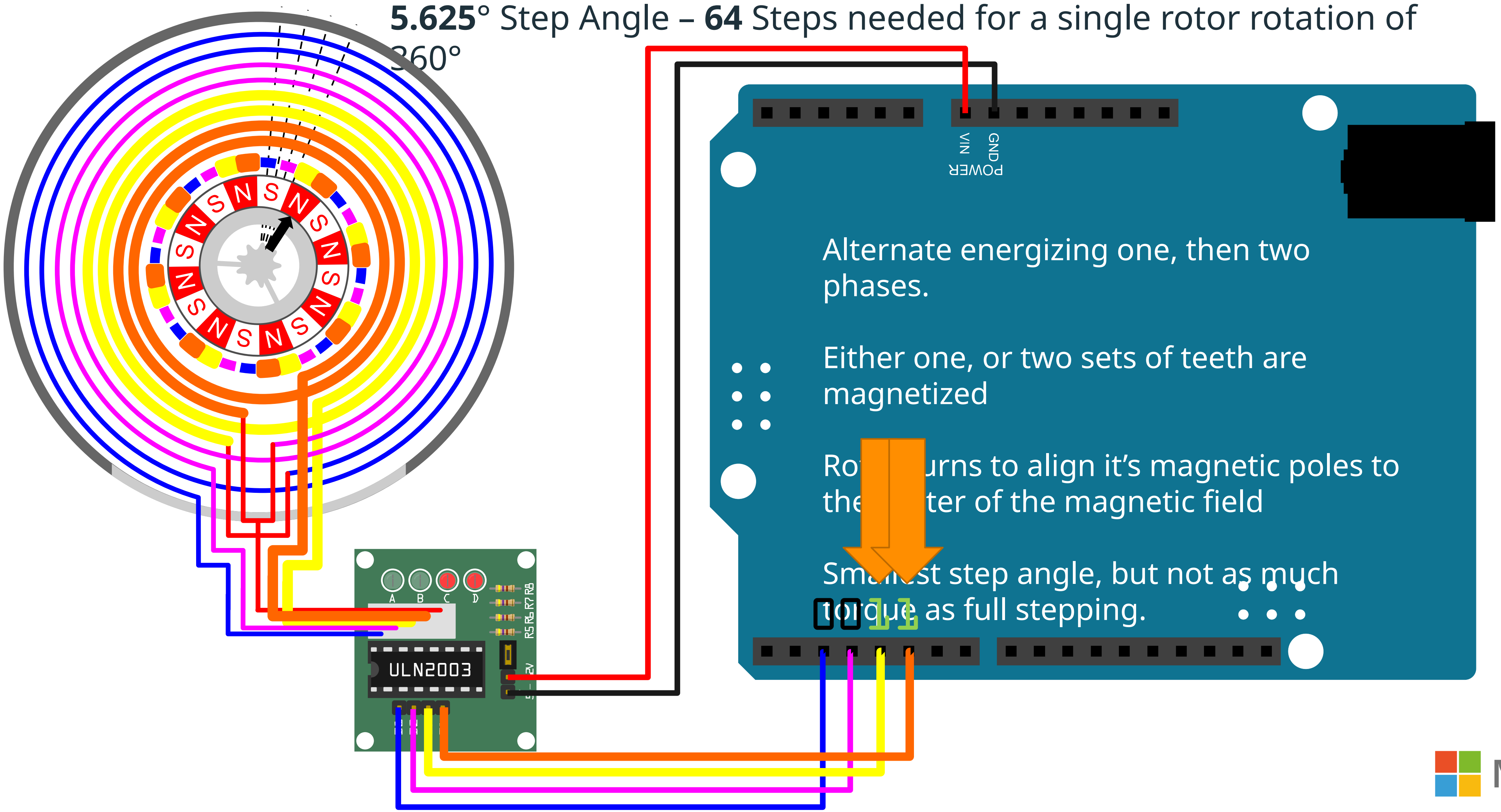
5.625° Step Angle – **64** Steps needed for a single rotor rotation of 360°

Alternate energizing one, then two phases.

Either one, or two sets of teeth are magnetized

Ro urns to align it's magnetic poles to the ter of the magnetic field

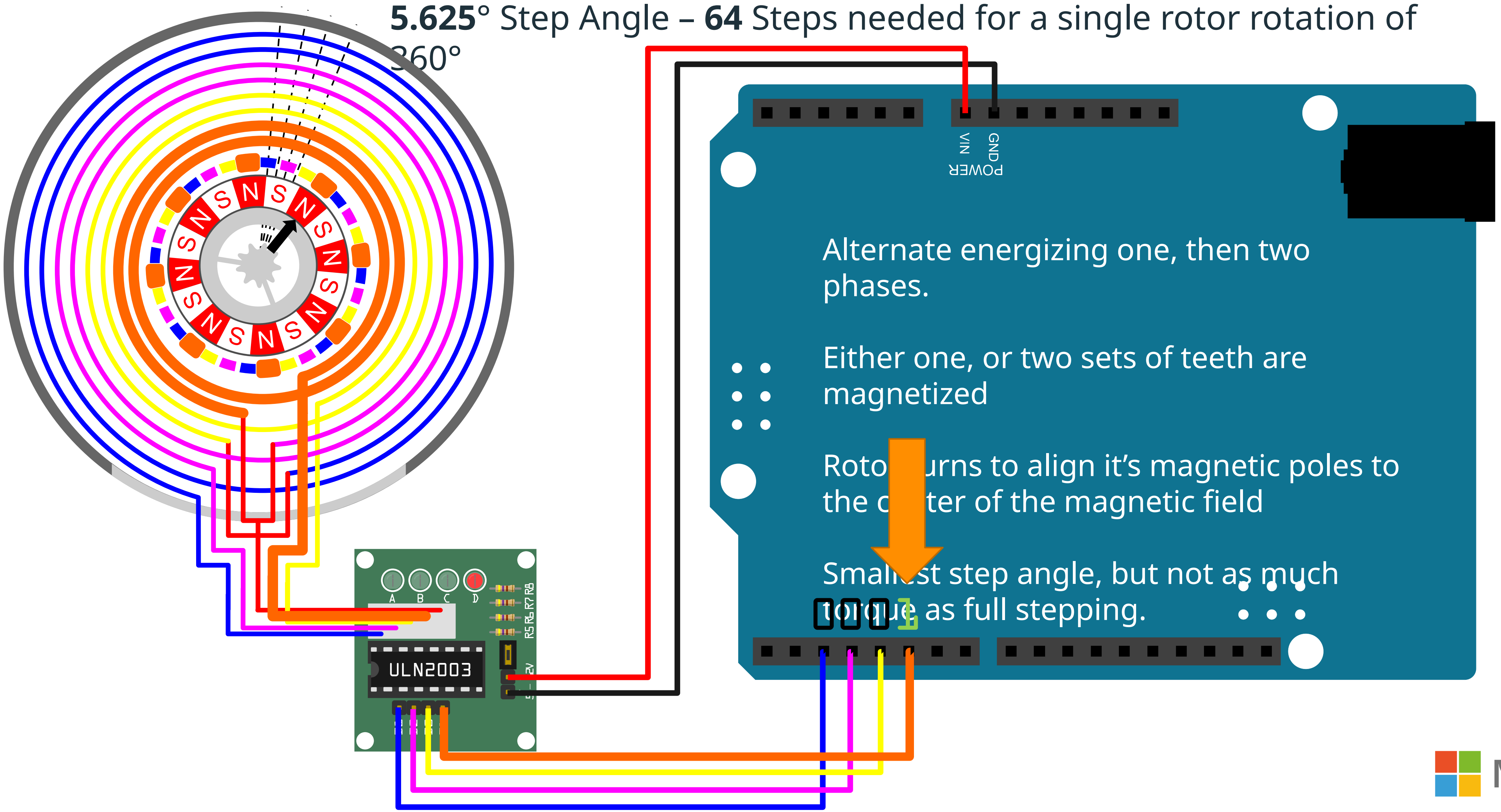
Sm est step angle, but not as much torque as full stepping.



Microsoft

HALF STEPPING – STEP 7

5.625° Step Angle – **64** Steps needed for a single rotor rotation of 360°



Alternate energizing one, then two phases.

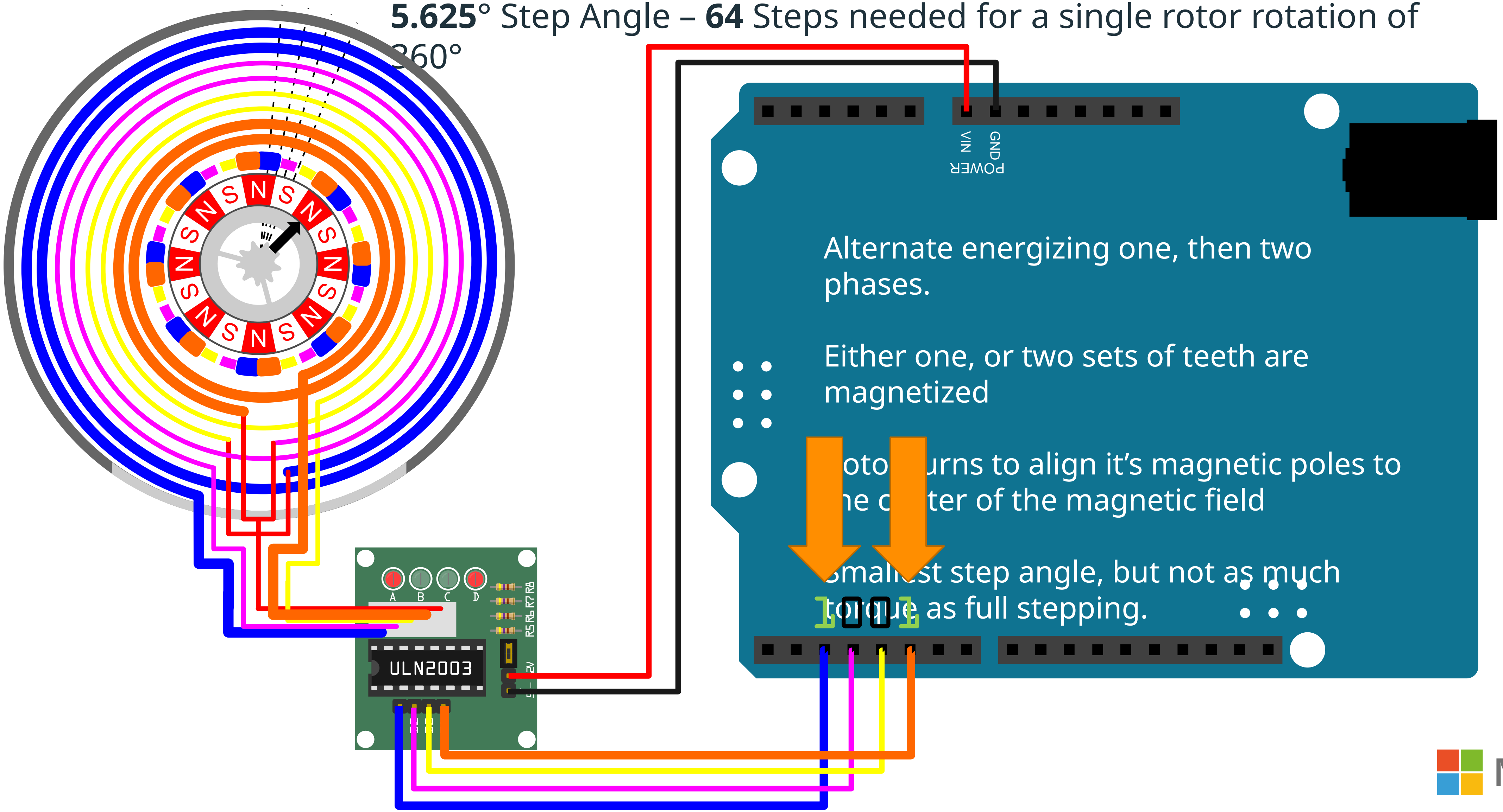
Either one, or two sets of teeth are magnetized

Rotor turns to align it's magnetic poles to the center of the magnetic field

Smallest step angle, but not as much torque as full stepping.

HALF STEPPING – STEP 8

5.625° Step Angle – **64** Steps needed for a single rotor rotation of 360°



Alternate energizing one, then two phases.

Either one, or two sets of teeth are magnetized

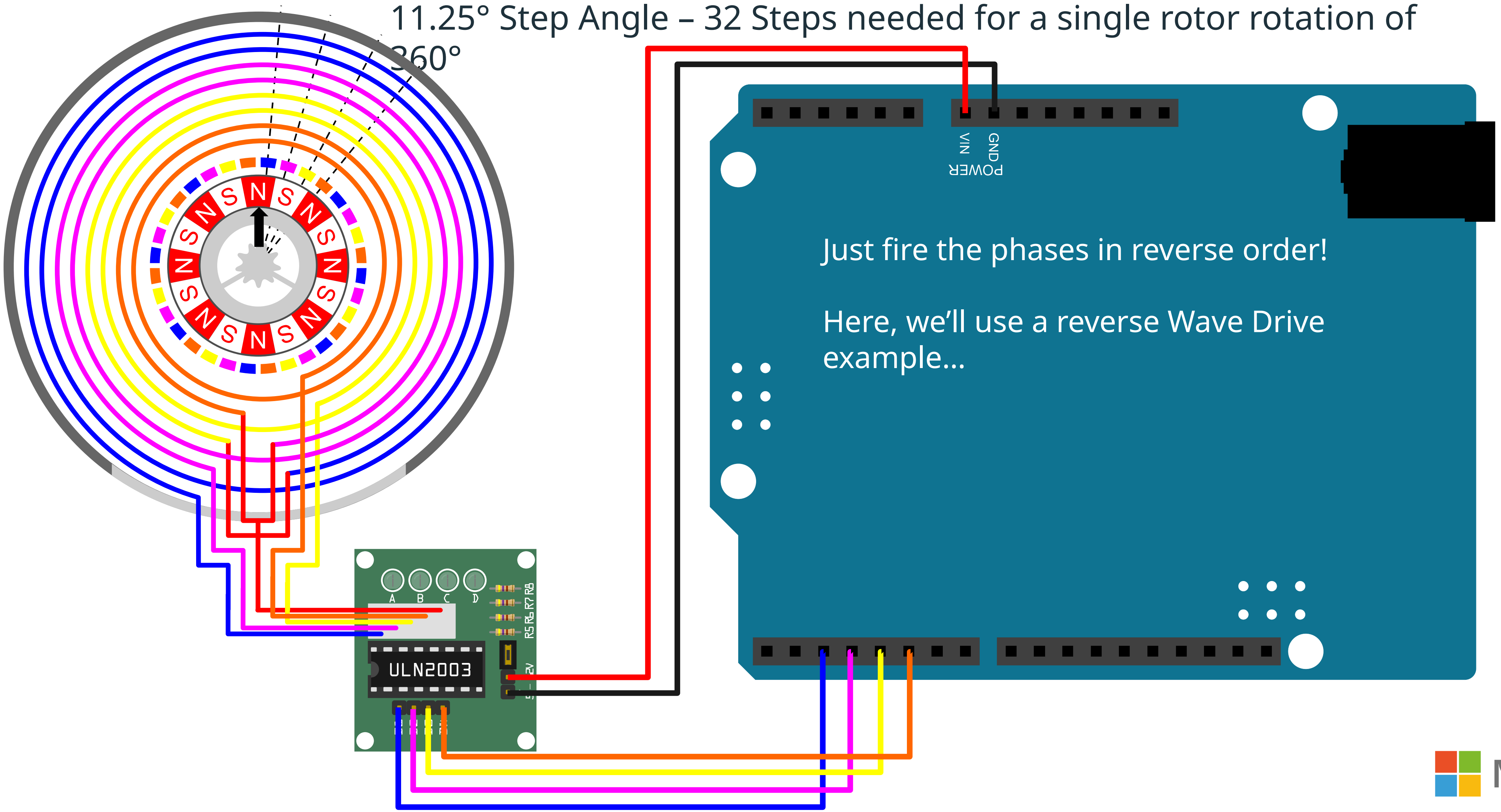
Rotor turns to align it's magnetic poles to the center of the magnetic field

Smallest step angle, but not as much torque as full stepping.

Microsoft

REVERSING THE DIRECTION

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°

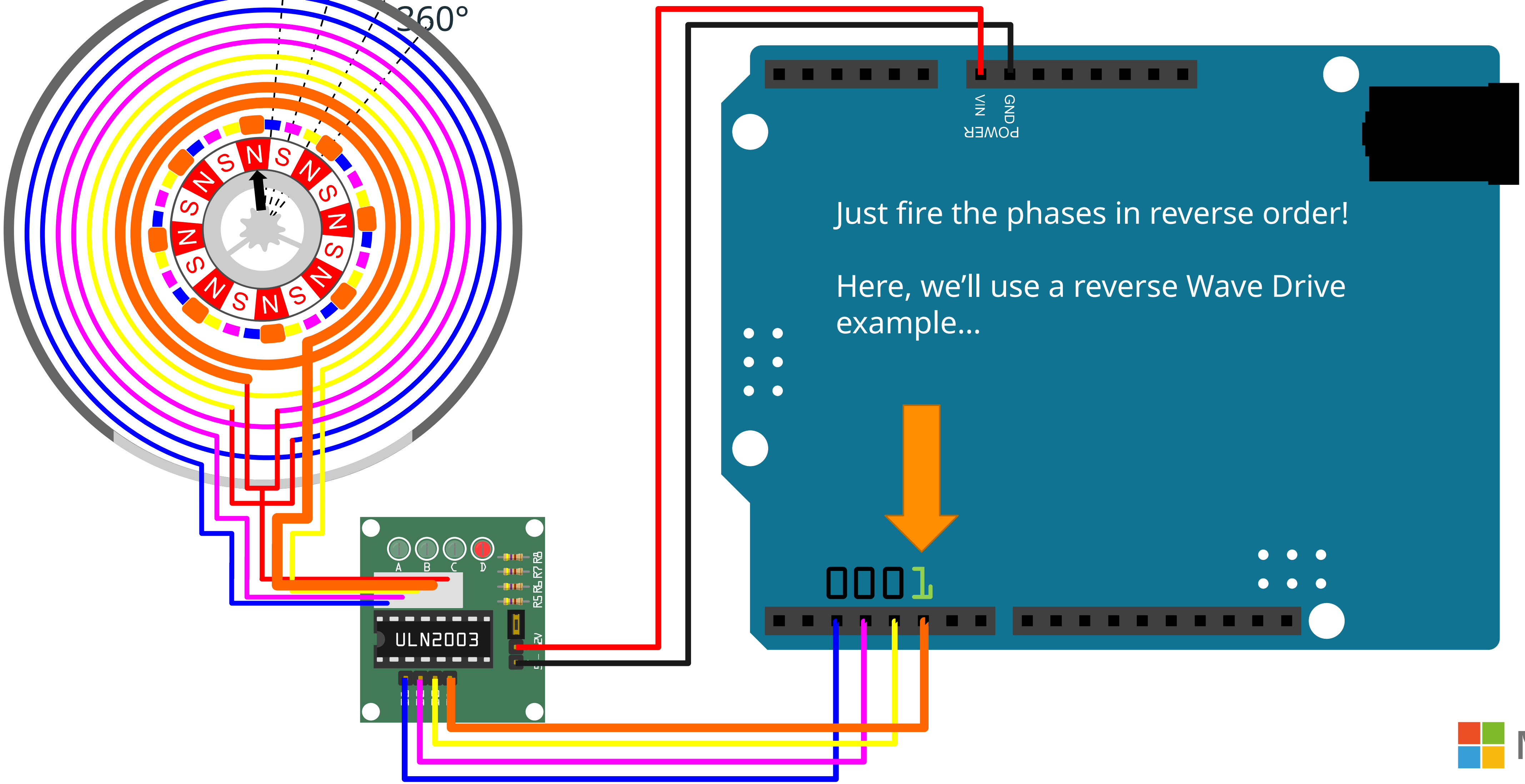


Just fire the phases in reverse order!

Here, we'll use a reverse Wave Drive example…

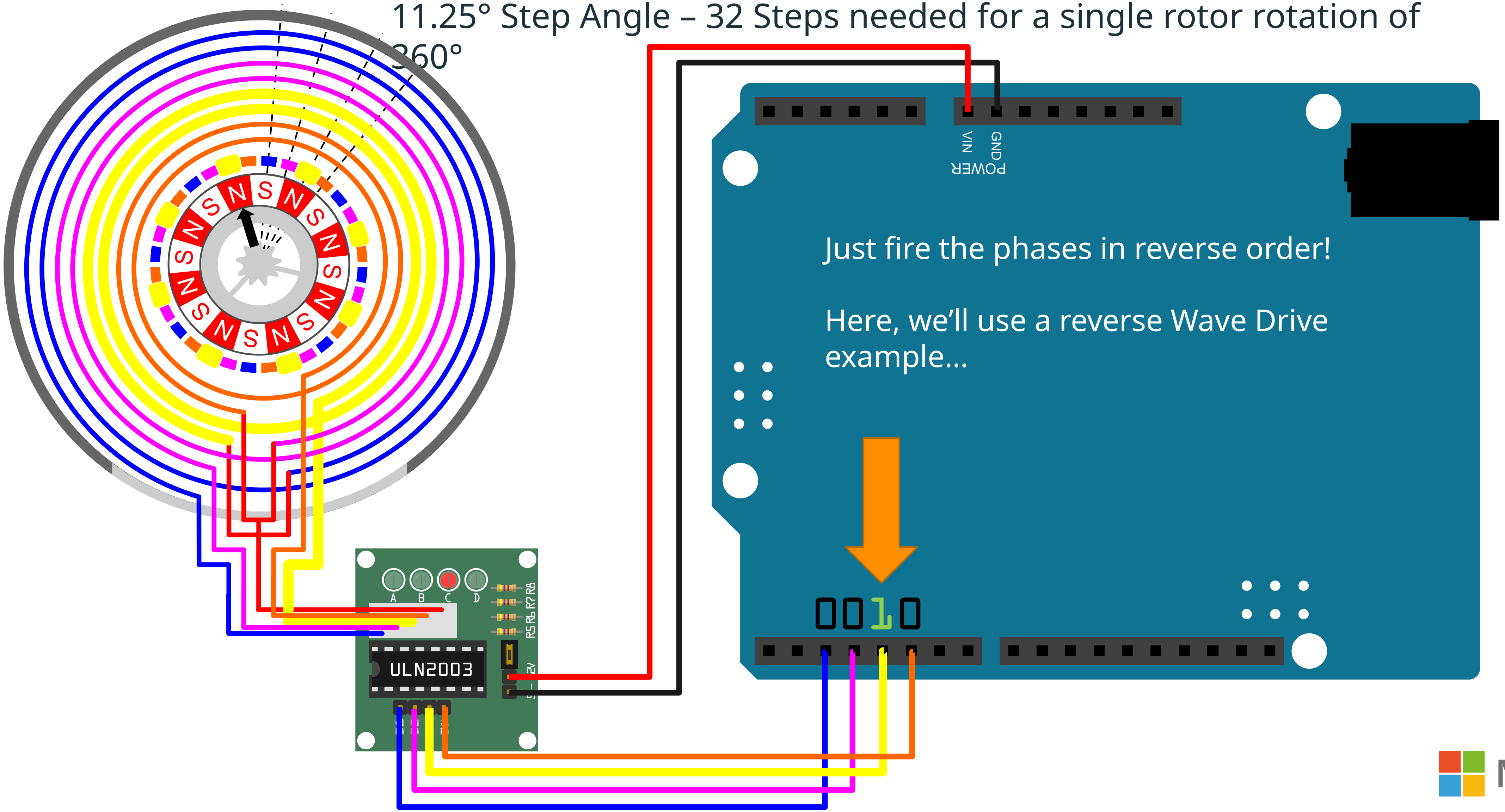
WAVE DRIVING IN REVERSE – STEP 1

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



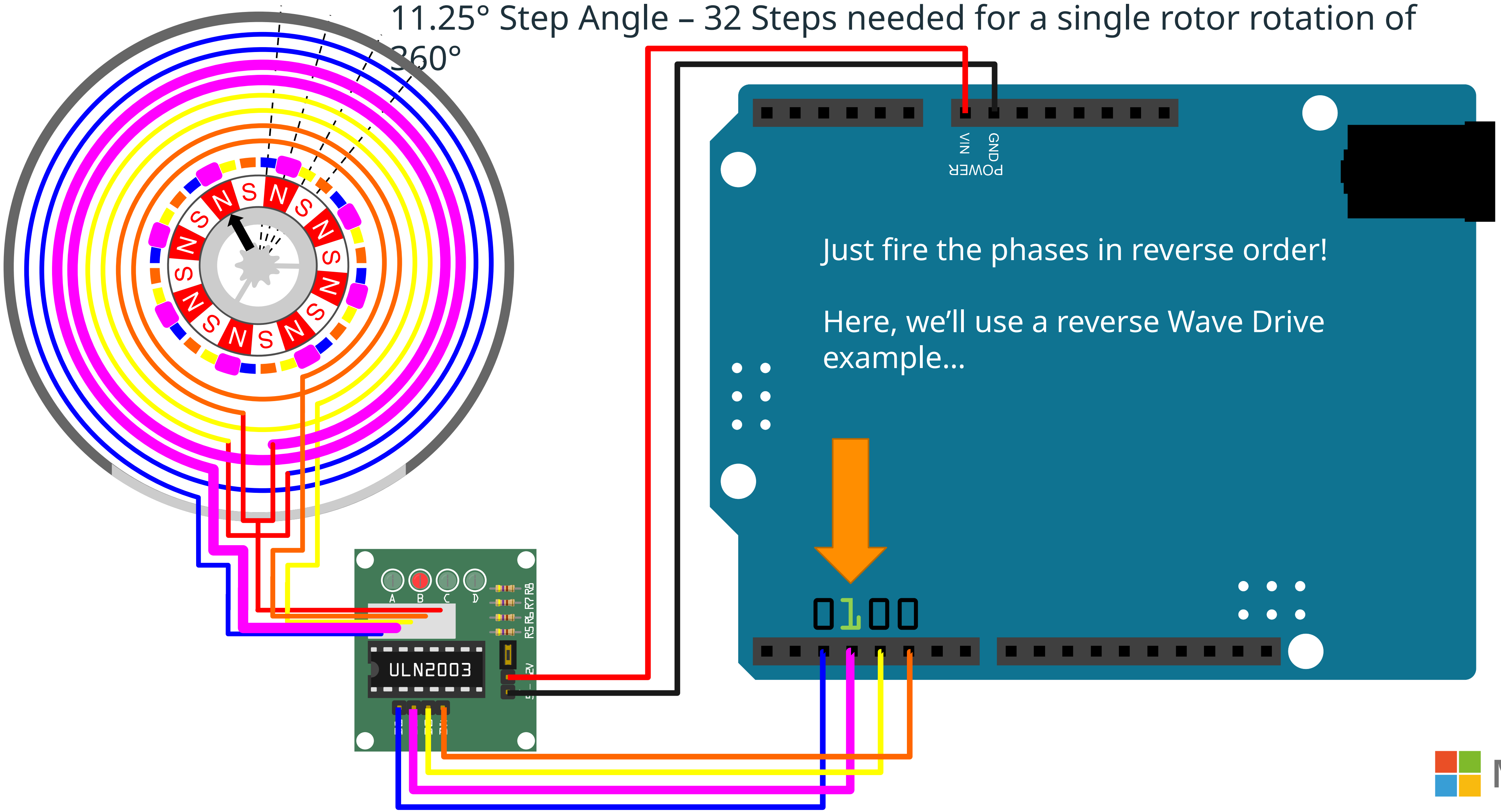
WAVE DRIVING IN REVERSE – STEP 2

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



WAVE DRIVING IN REVERSE – STEP 3

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



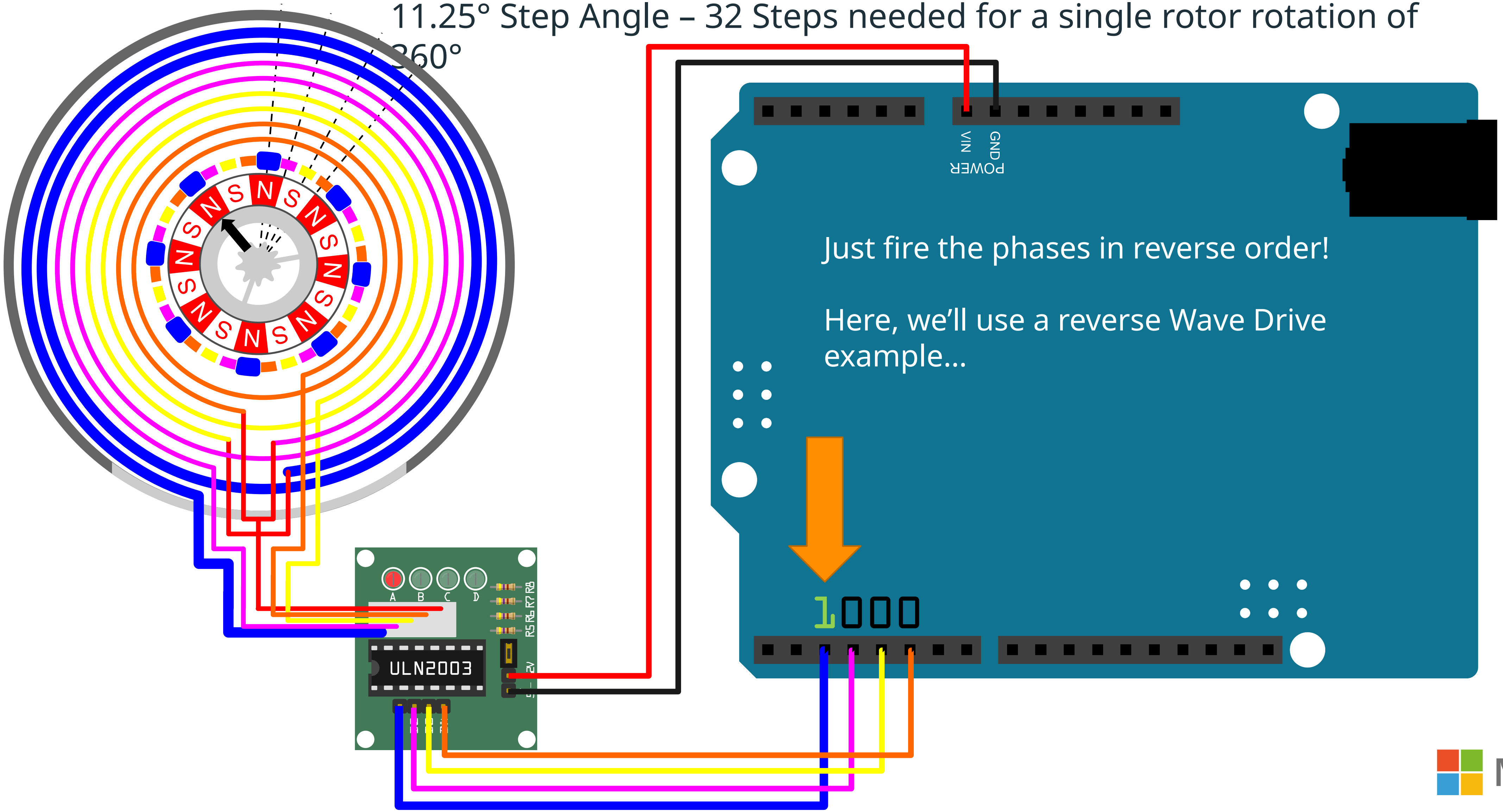
Just fire the phases in reverse order!

Here, we'll use a reverse Wave Drive example…

Microsoft

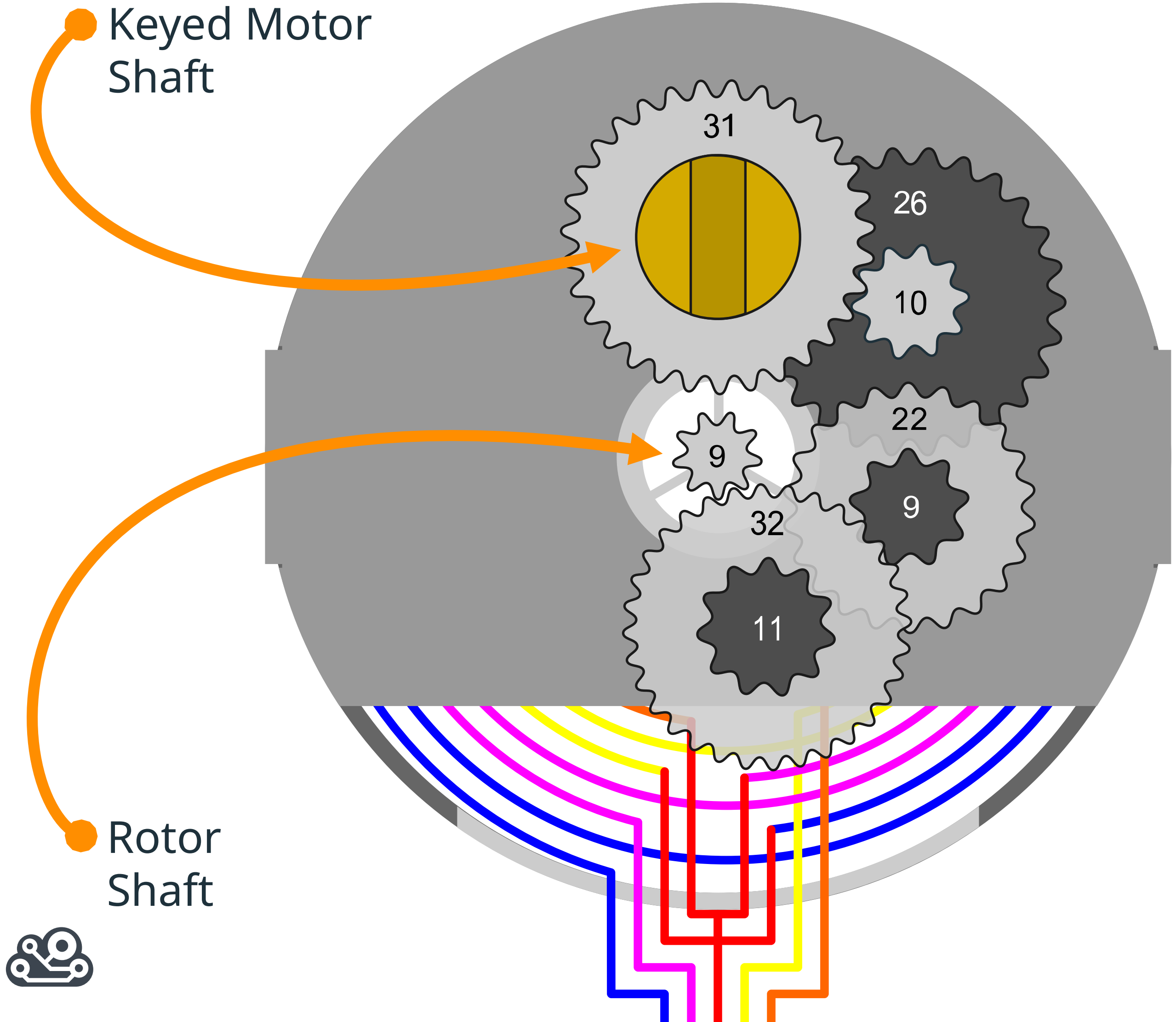
WAVE DRIVING IN REVERSE – STEP 4

11.25° Step Angle – 32 Steps needed for a single rotor rotation of 360°



GROKING THE GEARS

64:1 GEAR RATIO…



Gear Ratios:

32 / 9 = 3.555

22 / 11 = 2.000

26 / 9 = 2.888

31 / 10 = 3.100

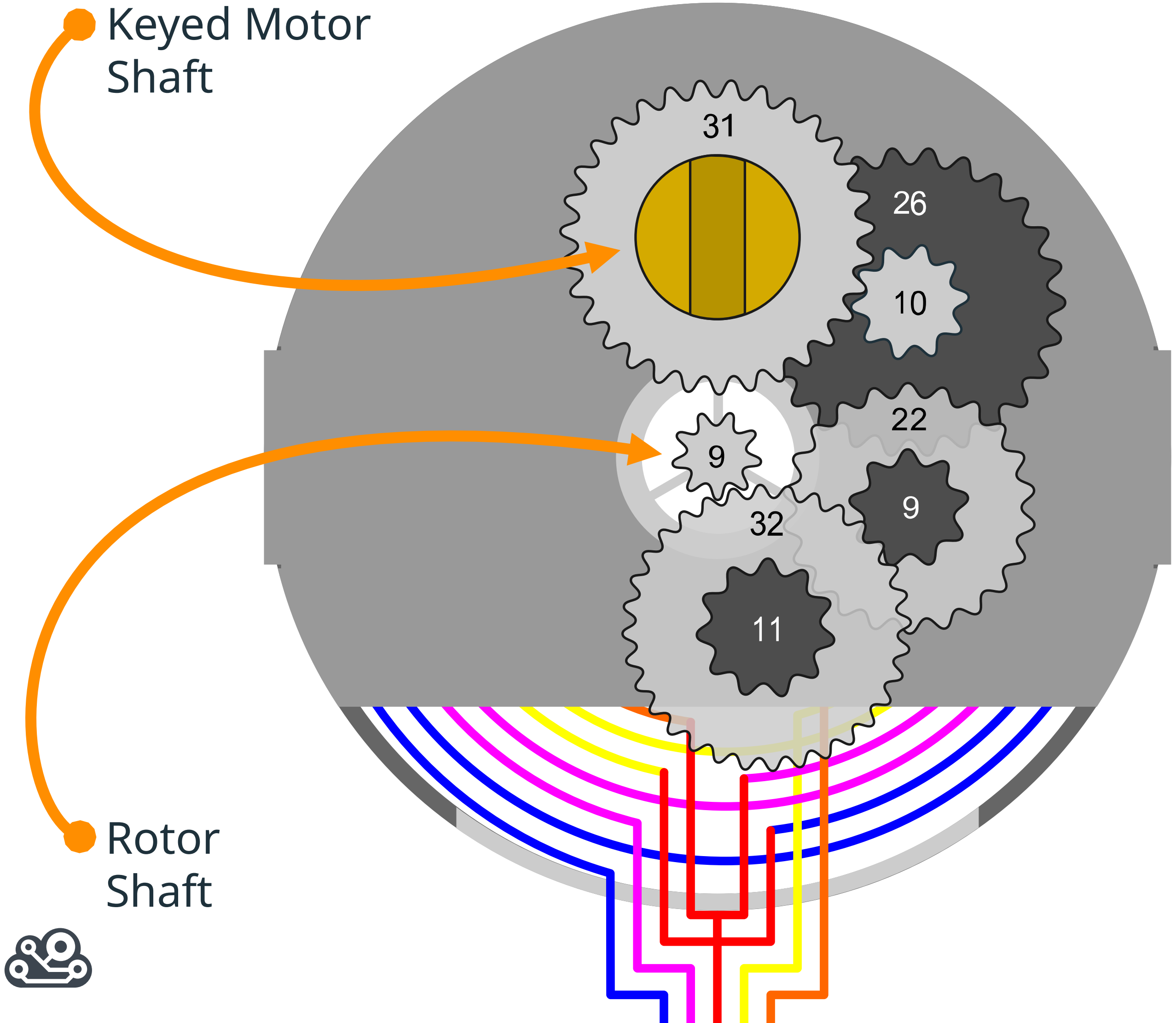
Multiply the gear ratios together:
3.555 x 2.000 x 2.888 x 3.100 = 63.65

Round 63.65 up:
64

That gives us a 64:1 gear ratio over all

It will take 64 full rotations of the "Rotor Shaft"
to get 1 rotation of the "Keyed Motor Shaft"

64:1 GEAR RATIO…



Gear Ratios:

32 / 9 = 3.555

22 / 11 = 2.000

26 / 9 = 2.888

31 / 10 = 3.100

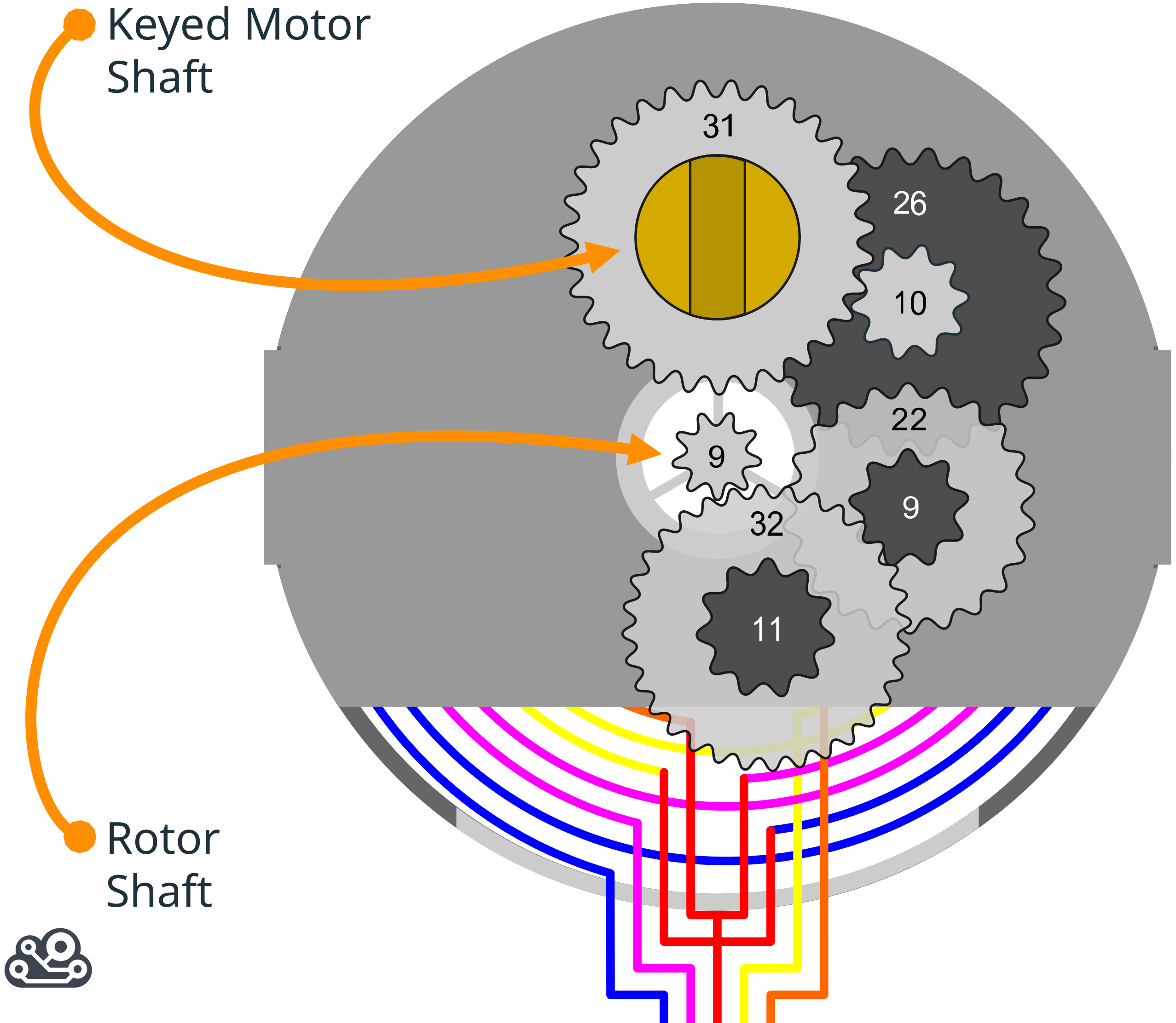
Multiply the gear ratios together:
3.555 x 2.000 x 2.888 x 3.100 = 63.65

Round 63.65 up:
64

That gives us a 64:1 gear ratio over all

It will take 64 full rotations of the "Rotor Shaft"
to get 1 rotation of the "Keyed Motor Shaft"

GEARS EFFECT ON MOTOR SHAFT STEPS



When full stepping, It takes **32** steps for the magnetic rotor inside the motor to complete a full 360° rotation, turning **11.25°** each step. Half stepping changes that to **64** steps of **5.625°** each for a full rotation.

The rotor shaft has a 64:1 ratio with the motor shaft though, so for the motor shaft, those numbers change to:

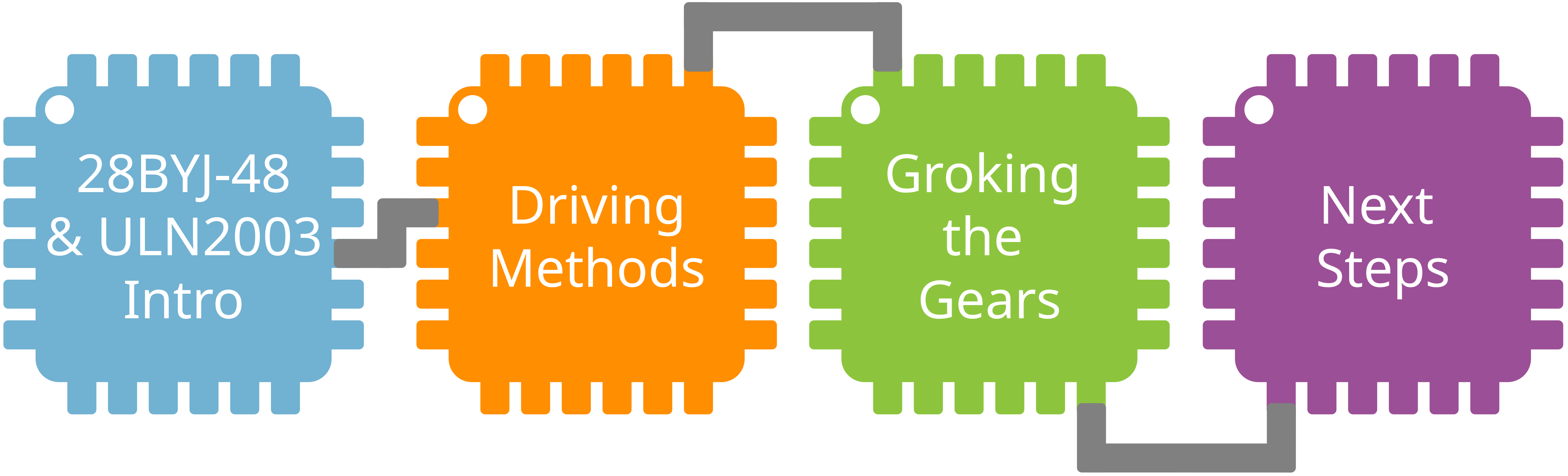
Full Stepping:
32*64 = **2048** Steps per motor shaft rotation
11.25° / 64 = **.18°** step angle

Half Stepping:
64*64 = **4096** Steps per motor shaft rotation
5.625° / 64 = **.09°** step angle

Microsoft

NEXT STEPS

REVIEW



28BYJ-48 & ULN2003 Intro
Driving Methods
Groking the Gears
Next Steps